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Breakdown of laser pulse energy for breakup of vascular calcium

Abstract

A catheter system (100) for treating a treatment site (106) within or adjacent to a blood vessel (108) includes a power source (124), a light guide (122) and a plasma target (242). In various embodiments, the light guide (122) receives power from the power source (124). The light guide (122) has a distal tip (244), and the light guide (122) emits light energy (243) in a direction away from the distal tip (244). The plasma target (242) is spaced apart from the distal tip (244) of the light guide (122) by a target gap distance (245). The plasma target (242) is configured to receive light energy (243) from the light guide (122) so that a plasma bubble (234) is generated at the plasma target (242). The power source (124) can be a laser and the light guide (122) can be an optical fiber. The catheter system (100) can also include an inflatable balloon (104) that encircles the distal tip (244) of the light guide (122). The plasma target (242) can be positioned within the inflatable balloon (104). The plasma target (242) can have a target face (1672) that receives the light energy (243) from the light guide (122). The plasma target (242) can be formed from one or more of tungsten, tantalum, platinum, molybdenum, niobium, iridium, magnesium oxide, beryllium oxide, tungsten carbide, titanium nitride, titanium carbonitride and titanium carbide.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4649924	12/1986	Taccardi	N/A	N/A
4699147	12/1986	Chilson et al.	N/A	N/A
4799479	12/1988	Spears	N/A	N/A
4850351	12/1988	Herman	N/A	N/A
4913142	12/1989	Kittrell et al.	N/A	N/A
4932954	12/1989	Wondrazek et al.	N/A	N/A
4955895	12/1989	Suglyama	N/A	N/A
4960108	12/1989	Reichel et al.	N/A	N/A
4994059	12/1990	Kosa et al.	N/A	N/A
4998930	12/1990	Lundahl	N/A	N/A
5034010	12/1990	Kittrell et al.	N/A	N/A
5041121	12/1990	Wondrazek et al.	N/A	N/A
5082343	12/1991	Coult et al.	N/A	N/A
5093877	12/1991	Aita et al.	N/A	N/A
5104391	12/1991	Ingle	N/A	N/A
5104392	12/1991	Kittrell et al.	N/A	N/A
5109452	12/1991	Selvin et al.	N/A	N/A
5116227	12/1991	Levy	N/A	N/A
5126165	12/1991	Akihama et al.	N/A	N/A
5152768	12/1991	Bhatta	N/A	N/A

5173049	12/1991	Levy	N/A	N/A
5176674	12/1992	Hofmann	N/A	N/A
5181921	12/1992	Makita et al.	N/A	N/A
5188632	12/1992	Goldenberg	N/A	N/A
5200838	12/1992	Nudelman	N/A	N/A
5269777	12/1992	Doiron	N/A	N/A
5290277	12/1993	Vercimak et al.	N/A	N/A
5324282	12/1993	Dodick	N/A	N/A
5328472	12/1993	Steinke et al.	N/A	N/A
5336184	12/1993	Teirstein	N/A	N/A
5363458	12/1993	Pan	N/A	N/A
5372138	12/1993	Crowley	N/A	N/A
5387225	12/1994	Euteneur	N/A	N/A
5400428	12/1994	Grace	N/A	N/A
5410797	12/1994	Steinke et al.	N/A	N/A
5422926	12/1994	Smith	N/A	N/A
5431647	12/1994	Purcell	N/A	N/A
5454809	12/1994	Janssen	N/A	N/A
5456680	12/1994	Taylor	N/A	N/A
5474537	12/1994	Solar	N/A	N/A
5509917	12/1995	Cecchetti	N/A	N/A
5540679	12/1995	Fram	N/A	N/A
5562657	12/1995	Griffin	N/A	N/A
5598494	12/1996	Behrmann et al.	N/A	N/A
5609606	12/1996	O'Boyle	N/A	N/A
5611807	12/1996	O'Boyle	N/A	N/A
5637877	12/1996	Sinofsky	N/A	N/A
5661829	12/1996	Zheng	N/A	N/A
5697377	12/1996	Wittkamp	N/A	N/A
5718241	12/1997	Ben-Haim et al.	N/A	N/A
5729583	12/1997	Tang	N/A	N/A
5764843	12/1997	Macken et al.	N/A	N/A
5772609	12/1997	Nguyen et al.	N/A	N/A
5860974	12/1998	Abele	N/A	N/A
5891135	12/1998	Jackson et al.	N/A	N/A
5906611	12/1998	Dodick et al.	N/A	N/A
5944697	12/1998	Benett et al.	N/A	N/A
6015404	12/1999	Altshuler	N/A	N/A
6080119	12/1999	Schwarze et al.	N/A	N/A
6123923	12/1999	Unger	N/A	N/A
6139510	12/1999	Palermo	N/A	N/A
6186963	12/2000	Schwarze et al.	N/A	N/A
6203537	12/2000	Adrian	N/A	N/A
6210404	12/2000	Shaddock	N/A	N/A
6339470	12/2001	Papademetriou et al.	N/A	N/A
6356575	12/2001	Fukumoto	N/A	N/A
6368318	12/2001	Visuri et al.	N/A	N/A
6423055	12/2001	Farr	N/A	N/A
6500174	12/2001	Maguire et al.	N/A	N/A
6514203	12/2002	Bukshpan	N/A	N/A

6514249	12/2002	Maguire	N/A	N/A
6524251	12/2002	Rabiner et al.	N/A	N/A
6538739	12/2002	Visuri et al.	N/A	N/A
6548010	12/2002	Stivland et al.	N/A	N/A
6560387	12/2002	Hehlen et al.	N/A	N/A
6607502	12/2002	Maguire et al.	N/A	N/A
6631220	12/2002	Liang et al.	N/A	N/A
6652547	12/2002	Rabiner et al.	N/A	N/A
6666834	12/2002	Restle et al.	N/A	N/A
6702781	12/2003	Reifart et al.	N/A	N/A
6773447	12/2003	Laguna	N/A	N/A
6824554	12/2003	Jang	N/A	N/A
6849994	12/2004	White et al.	N/A	N/A
6890317	12/2004	Gerdtts et al.	N/A	N/A
6947785	12/2004	Beatty et al.	N/A	N/A
6966890	12/2004	Coyle et al.	N/A	N/A
6978168	12/2004	Beatty et al.	N/A	N/A
6990370	12/2005	Beatty et al.	N/A	N/A
7273470	12/2006	Wantink	N/A	N/A
7309324	12/2006	Hayes et al.	N/A	N/A
7367967	12/2007	Eidenschink	N/A	N/A
7470240	12/2007	Schultheiss et al.	N/A	N/A
7539231	12/2008	Honea et al.	N/A	N/A
7569032	12/2008	Naimark et al.	N/A	N/A
7599588	12/2008	Eberle et al.	N/A	N/A
7641646	12/2009	Kennedy, II	N/A	N/A
7713260	12/2009	Lessard	N/A	N/A
7758572	12/2009	Weber et al.	N/A	N/A
7762984	12/2009	Kumoyama et al.	N/A	N/A
7810395	12/2009	Zhou	N/A	N/A
7850685	12/2009	Kunis et al.	N/A	N/A
7867178	12/2010	Simnacher	N/A	N/A
7909797	12/2010	Kennedy, II et al.	N/A	N/A
7967781	12/2010	Simpson et al.	N/A	N/A
7972299	12/2010	Carter	N/A	N/A
7985189	12/2010	Ogden et al.	N/A	N/A
8021328	12/2010	Lee	N/A	N/A
8029473	12/2010	Carter	N/A	N/A
8043256	12/2010	Hansen	N/A	N/A
8088121	12/2011	Nishide	N/A	N/A
8162859	12/2011	Schultheiss et al.	N/A	N/A
8166825	12/2011	Zhou	N/A	N/A
8192368	12/2011	Woodruff	N/A	N/A
8267886	12/2011	Ewing	N/A	N/A
8292913	12/2011	Warnack	N/A	N/A
8328820	12/2011	Diamant	N/A	N/A
8364235	12/2012	Kordis et al.	N/A	N/A
8372034	12/2012	Levit	N/A	N/A
8382738	12/2012	Simpson et al.	N/A	N/A
8414527	12/2012	Mallaby	N/A	N/A

8419613	12/2012	Saadat	N/A	N/A
8439890	12/2012	Beyar	N/A	N/A
8556813	12/2012	Cashman et al.	N/A	N/A
8556851	12/2012	Hirszowicz	N/A	N/A
8574247	12/2012	Adams et al.	N/A	N/A
8657814	12/2013	Werneth	N/A	N/A
8709075	12/2013	Adams et al.	N/A	N/A
8728091	12/2013	Hakala et al.	N/A	N/A
8734424	12/2013	Watanabe	N/A	N/A
8747416	12/2013	Hakala et al.	N/A	N/A
8784362	12/2013	Boutilette	N/A	N/A
8834510	12/2013	Wilson et al.	N/A	N/A
8888788	12/2013	Hakala et al.	N/A	N/A
8956371	12/2014	Hawkins et al.	N/A	N/A
8956374	12/2014	Hawkins et al.	N/A	N/A
8986339	12/2014	Warnack	N/A	N/A
8992817	12/2014	Stamberg	N/A	N/A
9005216	12/2014	Hakala et al.	N/A	N/A
9011462	12/2014	Adams et al.	N/A	N/A
9011463	12/2014	Adams et al.	N/A	N/A
9011511	12/2014	Gregorich	N/A	N/A
9044618	12/2014	Hawkins et al.	N/A	N/A
9044619	12/2014	Hawkins et al.	N/A	N/A
9072534	12/2014	Adams et al.	N/A	N/A
9089669	12/2014	Haslinger et al.	N/A	N/A
9131949	12/2014	Coleman et al.	N/A	N/A
9138249	12/2014	Adams et al.	N/A	N/A
9138260	12/2014	Miller et al.	N/A	N/A
9180280	12/2014	Hawkins et al.	N/A	N/A
9220521	12/2014	Hawkins et al.	N/A	N/A
9237984	12/2015	Hawkins et al.	N/A	N/A
9283359	12/2015	Pepper	N/A	N/A
9289132	12/2015	Ghaffari et al.	N/A	N/A
9289224	12/2015	Adams et al.	N/A	N/A
9320530	12/2015	Grace	N/A	N/A
9333000	12/2015	Hakala et al.	N/A	N/A
9339632	12/2015	Eidenschink et al.	N/A	N/A
9364645	12/2015	Erikawa	N/A	N/A
9375223	12/2015	Wallace	N/A	N/A
9421025	12/2015	Hawkins et al.	N/A	N/A
9433428	12/2015	Hakala et al.	N/A	N/A
9433745	12/2015	Cully	N/A	N/A
9504809	12/2015	Bo	N/A	N/A
9510887	12/2015	Burnett	N/A	N/A
9522012	12/2015	Adams	N/A	N/A
9554815	12/2016	Adams et al.	N/A	N/A
9555267	12/2016	Ein-gal	N/A	N/A
9566209	12/2016	Katragadda et al.	N/A	N/A
9579114	12/2016	Mantell et al.	N/A	N/A
9579492	12/2016	Simpson	N/A	N/A

9585684	12/2016	Nita et al.	N/A	N/A
9592328	12/2016	Jeevanandam	N/A	N/A
9629567	12/2016	Porath et al.	N/A	N/A
9642673	12/2016	Adams	N/A	N/A
9662069	12/2016	De Graff et al.	N/A	N/A
9687166	12/2016	Subramaniam	N/A	N/A
9730715	12/2016	Adams	N/A	N/A
9737361	12/2016	Magana	N/A	N/A
9764142	12/2016	Imran	N/A	N/A
9782570	12/2016	Hirszowicz	N/A	N/A
9814476	12/2016	Adams et al.	N/A	N/A
9833348	12/2016	Jordan et al.	N/A	N/A
9839764	12/2016	Chouinard	N/A	N/A
9861377	12/2017	Mantell et al.	N/A	N/A
9867629	12/2017	Hawkins et al.	N/A	N/A
9878135	12/2017	Holzapfel et al.	N/A	N/A
9894756	12/2017	Weinkam et al.	N/A	N/A
9901704	12/2017	Appling	N/A	N/A
9955946	12/2017	Miller et al.	N/A	N/A
9974963	12/2017	Imran	N/A	N/A
9974970	12/2017	Nuta et al.	N/A	N/A
9993292	12/2017	Adams et al.	N/A	N/A
10039561	12/2017	Adams et al.	N/A	N/A
10076384	12/2017	Kasprzyk	N/A	N/A
10086175	12/2017	Torres et al.	N/A	N/A
10124153	12/2017	Feig	N/A	N/A
10136829	12/2017	Deno et al.	N/A	N/A
10149690	12/2017	Hawkins et al.	N/A	N/A
10159505	12/2017	Hakala et al.	N/A	N/A
10194994	12/2018	Deno et al.	N/A	N/A
10201387	12/2018	Grace et al.	N/A	N/A
10206698	12/2018	Hakala et al.	N/A	N/A
10226265	12/2018	Ku et al.	N/A	N/A
10245410	12/2018	Aggerholm	N/A	N/A
10357264	12/2018	Kat-Kuoy	N/A	N/A
10405923	12/2018	Yu et al.	N/A	N/A
10406031	12/2018	Thyzel	N/A	N/A
10406318	12/2018	Williams	N/A	N/A
10420569	12/2018	Adams	N/A	N/A
10439791	12/2018	Kalhan	N/A	N/A
10441300	12/2018	Hawkins	N/A	N/A
10449339	12/2018	Wilson et al.	N/A	N/A
10463430	12/2018	Dick	N/A	N/A
10478202	12/2018	Adams et al.	N/A	N/A
10517620	12/2018	Adams	N/A	N/A
10517621	12/2018	Hakala et al.	N/A	N/A
10537287	12/2019	Braido et al.	N/A	N/A
10555744	12/2019	Nguyen et al.	N/A	N/A
10561428	12/2019	Eggert et al.	N/A	N/A
10583277	12/2019	Rundquist	N/A	N/A

10589073	12/2019	Mallaby	N/A	N/A
10617850	12/2019	Tal	N/A	N/A
10646240	12/2019	Betelia et al.	N/A	N/A
10668245	12/2019	Kanae	N/A	N/A
10682178	12/2019	Adams et al.	N/A	N/A
10695531	12/2019	Suzuki	N/A	N/A
10702293	12/2019	Adams et al.	N/A	N/A
10709462	12/2019	Nguyen et al.	N/A	N/A
10709872	12/2019	Alvarez et al.	N/A	N/A
10758255	12/2019	Adams	N/A	N/A
10797684	12/2019	Benz et al.	N/A	N/A
10799688	12/2019	Calhoun	N/A	N/A
10842567	12/2019	Grace et al.	N/A	N/A
10850075	12/2019	Tarunaga	N/A	N/A
10857329	12/2019	Davies	N/A	N/A
10933225	12/2020	Campbell	N/A	N/A
10959743	12/2020	Adams et al.	N/A	N/A
10966737	12/2020	Nguyen	N/A	N/A
10967156	12/2020	Gulachenski	N/A	N/A
10973538	12/2020	Hakala et al.	N/A	N/A
10980987	12/2020	Tarunaga	N/A	N/A
11000299	12/2020	Hawkins et al.	N/A	N/A
11020135	12/2020	Hawkins	N/A	N/A
11026707	12/2020	Ku et al.	N/A	N/A
11058492	12/2020	Grace et al.	N/A	N/A
11076874	12/2020	Hakala et al.	N/A	N/A
11116939	12/2020	Jamous et al.	N/A	N/A
11141131	12/2020	Stigall	N/A	N/A
11207493	12/2020	Suzuki et al.	N/A	N/A
11213661	12/2021	Spindler	N/A	N/A
11229772	12/2021	Nita	N/A	N/A
11229776	12/2021	Kugler et al.	N/A	N/A
11246659	12/2021	Grace et al.	N/A	N/A
11253681	12/2021	Williams	N/A	N/A
11389171	12/2021	Goldsmith	N/A	N/A
11484327	12/2021	Anderson et al.	N/A	N/A
11633200	12/2022	Anderson et al.	N/A	N/A
11779363	12/2022	Vo	N/A	N/A
11826530	12/2022	Suzuki	N/A	N/A
11839391	12/2022	Schultheis et al.	N/A	N/A
11911054	12/2023	Singla	N/A	N/A
11911056	12/2023	Anderson et al.	N/A	N/A
11918285	12/2023	Sun et al.	N/A	N/A
11944331	12/2023	Anderson et al.	N/A	N/A
11950793	12/2023	Nguyen	N/A	N/A
12035932	12/2023	Nunes	N/A	N/A
12144516	12/2023	Betelia	N/A	N/A
2001/0016761	12/2000	Rudie	N/A	N/A
2001/0018569	12/2000	Erbel	N/A	N/A
2001/0020164	12/2000	Papademetriou	N/A	N/A

2001/0049464	12/2000	Ganz	N/A	N/A
2001/0051784	12/2000	Brisken	N/A	N/A
2002/0045811	12/2001	Kittrell et al.	N/A	N/A
2002/0052621	12/2001	Fried et al.	N/A	N/A
2002/0065512	12/2001	Fjield et al.	N/A	N/A
2002/0082553	12/2001	Duchamp	N/A	N/A
2002/0183620	12/2001	Tearney	N/A	N/A
2002/0183729	12/2001	Farr et al.	N/A	N/A
2002/0188204	12/2001	McNamara et al.	N/A	N/A
2003/0009157	12/2002	Levine et al.	N/A	N/A
2003/0050632	12/2002	Fjield et al.	N/A	N/A
2003/0065316	12/2002	Levine et al.	N/A	N/A
2003/0114901	12/2002	Loeb et al.	N/A	N/A
2003/0125719	12/2002	Furnish	N/A	N/A
2003/0144654	12/2002	Hilal	N/A	N/A
2003/0176873	12/2002	Chernenko et al.	N/A	N/A
2004/0002677	12/2003	Gentsler	N/A	N/A
2004/0024349	12/2003	Flock et al.	N/A	N/A
2004/0073251	12/2003	Weber	N/A	N/A
2004/0097996	12/2003	Rabiner	N/A	N/A
2004/0133254	12/2003	Sterzer et al.	N/A	N/A
2004/0162508	12/2003	Uebelacker	N/A	N/A
2004/0210278	12/2003	Boll	N/A	N/A
2004/0243119	12/2003	Lane et al.	N/A	N/A
2004/0249401	12/2003	Rabiner	N/A	N/A
2004/0254570	12/2003	Hadsjicostis	N/A	N/A
2005/0010095	12/2004	Stewart et al.	N/A	N/A
2005/0021013	12/2004	Visuri	N/A	N/A
2005/0080396	12/2004	Rontal	N/A	N/A
2005/0113722	12/2004	Schultheiss	N/A	N/A
2005/0171437	12/2004	Carberry	N/A	N/A
2005/0171527	12/2004	Bhola	N/A	N/A
2005/0251131	12/2004	Lesh	N/A	N/A
2005/0259319	12/2004	Brooker	N/A	N/A
2005/0273014	12/2004	Gianchandani et al.	N/A	N/A
2005/0277839	12/2004	Alderman et al.	N/A	N/A
2006/0033241	12/2005	Schewe et al.	N/A	N/A
2006/0084966	12/2005	Maguire et al.	N/A	N/A
2006/0098921	12/2005	Benaron et al.	N/A	N/A
2006/0142703	12/2005	Carter	N/A	N/A
2006/0190022	12/2005	Beyar et al.	N/A	N/A
2006/0200039	12/2005	Brockway et al.	N/A	N/A
2006/0221528	12/2005	Li et al.	N/A	N/A
2006/0241524	12/2005	Lee et al.	N/A	N/A
2006/0241572	12/2005	Zhou	N/A	N/A
2006/0241733	12/2005	Zhang et al.	N/A	N/A
2006/0270976	12/2005	Savage et al.	N/A	N/A
2007/0027524	12/2006	Johnson	N/A	N/A
2007/0043340	12/2006	Thyzel	N/A	N/A
2007/0060990	12/2006	Satake	N/A	N/A

2007/0088380	12/2006	Hirszowicz et al.	N/A	N/A
2007/0118057	12/2006	Ein-gal	N/A	N/A
2007/0142779	12/2006	Duane	N/A	N/A
2007/0142819	12/2006	El-Nounou et al.	N/A	N/A
2007/0142821	12/2006	Hennessy et al.	N/A	N/A
2007/0142856	12/2006	Jang	N/A	N/A
2007/0179496	12/2006	Swoyer	N/A	N/A
2007/0239082	12/2006	Schultheiss et al.	N/A	N/A
2007/0255270	12/2006	Carney	N/A	N/A
2007/0264353	12/2006	Myntti et al.	N/A	N/A
2007/0270897	12/2006	Skerven	N/A	N/A
2007/0280311	12/2006	Hofmann	N/A	N/A
2007/0299392	12/2006	Beyar et al.	N/A	N/A
2008/0033519	12/2007	Burwell	N/A	N/A
2008/0081950	12/2007	Koenig et al.	N/A	N/A
2008/0086118	12/2007	Lai	N/A	N/A
2008/0095714	12/2007	Castella et al.	N/A	N/A
2008/0097251	12/2007	Babaev	N/A	N/A
2008/0108867	12/2007	Zhou	N/A	N/A
2008/0114341	12/2007	Thyzel	N/A	N/A
2008/0132810	12/2007	Scoseria et al.	N/A	N/A
2008/0175539	12/2007	Brown	N/A	N/A
2008/0195088	12/2007	Farr et al.	N/A	N/A
2008/0214891	12/2007	Slenker et al.	N/A	N/A
2008/0221550	12/2007	Lee	N/A	N/A
2008/0281157	12/2007	Miyagi et al.	N/A	N/A
2008/0296152	12/2007	Voss	N/A	N/A
2008/0319356	12/2007	Cain et al.	N/A	N/A
2009/0036803	12/2008	Warlick et al.	N/A	N/A
2009/0043300	12/2008	Reitmayer et al.	N/A	N/A
2009/0054881	12/2008	Krespi	N/A	N/A
2009/0097806	12/2008	Viellerobe et al.	N/A	N/A
2009/0125007	12/2008	Splinter	N/A	N/A
2009/0131921	12/2008	Kurtz et al.	N/A	N/A
2009/0192495	12/2008	Ostrovsky et al.	N/A	N/A
2009/0240242	12/2008	Neuberger	N/A	N/A
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2009/0296751	12/2008	Kewitsch et al.	N/A	N/A
2009/0299327	12/2008	Tilson et al.	N/A	N/A
2009/0306533	12/2008	Rousche	N/A	N/A
2009/0312768	12/2008	Hawkins et al.	N/A	N/A
2010/0016862	12/2009	Hawkins et al.	N/A	N/A
2010/0036238	12/2009	Neidert	N/A	N/A
2010/0036294	12/2009	Mantell et al.	N/A	N/A
2010/0063491	12/2009	Verhagen	N/A	N/A
2010/0094209	12/2009	Drasler et al.	N/A	N/A
2010/0114020	12/2009	Hawkins et al.	N/A	N/A
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2010/0125268	12/2009	Gustus et al.	N/A	N/A
2010/0160838	12/2009	Krespi	N/A	N/A
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2010/0168862	12/2009	Edie et al.	N/A	N/A
2010/0179632	12/2009	Bruszewski et al.	N/A	N/A
2010/0191089	12/2009	Stebler et al.	N/A	N/A
2010/0198114	12/2009	Novak et al.	N/A	N/A
2010/0199773	12/2009	Zhou	N/A	N/A
2010/0222786	12/2009	Kassab	N/A	N/A
2010/0234875	12/2009	Allex et al.	N/A	N/A
2010/0256535	12/2009	Novak et al.	N/A	N/A
2010/0265733	12/2009	O'Leary	N/A	N/A
2010/0286791	12/2009	Goldsmith	N/A	N/A
2010/0316333	12/2009	Luther	N/A	N/A
2011/0034832	12/2010	Cioanta et al.	N/A	N/A
2011/0059415	12/2010	Kasenbacher	N/A	N/A
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2011/0118634	12/2010	Golan	N/A	N/A
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2011/0184244	12/2010	Kagaya et al.	N/A	N/A
2011/0208185	12/2010	Diamant et al.	N/A	N/A
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2011/0245740	12/2010	Novak et al.	N/A	N/A
2011/0257641	12/2010	Hastings et al.	N/A	N/A
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2011/0275990	12/2010	Besser et al.	N/A	N/A
2011/0306956	12/2010	Islam	N/A	N/A
2012/0064141	12/2011	Andreacchi et al.	N/A	N/A
2012/0071715	12/2011	Beyar et al.	N/A	N/A
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2012/0089132	12/2011	Dick et al.	N/A	N/A
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2012/0095461	12/2011	Herscher et al.	N/A	N/A
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2012/0296367	12/2011	Grovender et al.	N/A	N/A
2012/0323211	12/2011	Ogle	N/A	N/A
2012/0330293	12/2011	Arai	N/A	N/A
2013/0030431	12/2012	Adams	N/A	N/A

2013/0030447	12/2012	Adams	N/A	N/A
2013/0041355	12/2012	Heeren et al.	N/A	N/A
2013/0046207	12/2012	Capelli	N/A	N/A
2013/0046293	12/2012	Arai et al.	N/A	N/A
2013/0053762	12/2012	Rontal et al.	N/A	N/A
2013/0060234	12/2012	Besser	N/A	N/A
2013/0110003	12/2012	Surti	N/A	N/A
2013/0116714	12/2012	Adams et al.	N/A	N/A
2013/0165764	12/2012	Scheuermann	N/A	N/A
2013/0190803	12/2012	Angel et al.	N/A	N/A
2013/0197614	12/2012	Gustus	N/A	N/A
2013/0218054	12/2012	Sverdlik et al.	N/A	N/A
2013/0226131	12/2012	Bacino et al.	N/A	N/A
2013/0253466	12/2012	Campbell	N/A	N/A
2013/0274726	12/2012	Takayama	N/A	N/A
2013/0345617	12/2012	Wallace	N/A	N/A
2014/0005576	12/2013	Adams	N/A	N/A
2014/0005706	12/2013	Gelfand et al.	N/A	N/A
2014/0012186	12/2013	Thyzel	N/A	N/A
2014/0039002	12/2013	Diodone et al.	N/A	N/A
2014/0039358	12/2013	Zhou et al.	N/A	N/A
2014/0039513	12/2013	Hakala	N/A	N/A
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2014/0046353	12/2013	Adams	N/A	N/A
2014/0052146	12/2013	Curtis et al.	N/A	N/A
2014/0052147	12/2013	Hakala et al.	N/A	N/A
2014/0058294	12/2013	Gross et al.	N/A	N/A
2014/0074111	12/2013	Hakala	N/A	N/A
2014/0153087	12/2013	Hutchings et al.	N/A	N/A
2014/0155990	12/2013	Nyuli	N/A	N/A
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2015/0005576	12/2014	Diodone et al.	N/A	N/A
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2015/0057648	12/2014	Swift et al.	N/A	N/A
2015/0071591	12/2014	Chen	N/A	N/A
2015/0073430	12/2014	Hakala et al.	N/A	N/A
2015/0080875	12/2014	Kasprzyk et al.	N/A	N/A

2015/0100048	12/2014	Hiereth et al.	N/A	N/A
2015/0105715	12/2014	Pikus et al.	N/A	N/A
2015/0119870	12/2014	Rudie	N/A	N/A
2015/0126990	12/2014	Sharma	N/A	N/A
2015/0141764	12/2014	Harks et al.	N/A	N/A
2015/0250542	12/2014	Islam	N/A	N/A
2015/0276689	12/2014	Watanabe et al.	N/A	N/A
2015/0313732	12/2014	Fulton, III	N/A	N/A
2015/0320432	12/2014	Adams	N/A	N/A
2015/0342678	12/2014	Deladurantaye et al.	N/A	N/A
2015/0359432	12/2014	Ehrenreich	N/A	N/A
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2016/0008016	12/2015	Cioanta et al.	N/A	N/A
2016/0016016	12/2015	Taylor et al.	N/A	N/A
2016/0018602	12/2015	Govari et al.	N/A	N/A
2016/0022294	12/2015	Cioanta et al.	N/A	N/A
2016/0038087	12/2015	Hunter	N/A	N/A
2016/0095610	12/2015	Lipowski et al.	N/A	N/A
2016/0135828	12/2015	Hawkins et al.	N/A	N/A
2016/0135891	12/2015	Feldman	N/A	N/A
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2016/0151639	12/2015	Scharf et al.	N/A	N/A
2016/0183819	12/2015	Burnett	N/A	N/A
2016/0183957	12/2015	Hakala et al.	N/A	N/A
2016/0184020	12/2015	Kowalewski et al.	N/A	N/A
2016/0184022	12/2015	Grace et al.	N/A	N/A
2016/0184023	12/2015	Grace et al.	N/A	N/A
2016/0184526	12/2015	Beyar	N/A	N/A
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2016/0228187	12/2015	Gross	N/A	N/A
2016/0234534	12/2015	Kitahara et al.	N/A	N/A
2016/0262784	12/2015	Grace et al.	N/A	N/A
2016/0270806	12/2015	Wallace	N/A	N/A
2016/0302762	12/2015	Stigall et al.	N/A	N/A
2016/0324564	12/2015	Gerlach et al.	N/A	N/A
2016/0331389	12/2015	Hakala et al.	N/A	N/A
2016/0339204	12/2015	Williams	N/A	N/A
2016/0367274	12/2015	Wallace	N/A	N/A
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2017/0049463	12/2016	Popovic et al.	N/A	N/A
2017/0056035	12/2016	Adams	N/A	N/A
2017/0086867	12/2016	Adams	N/A	N/A
2017/0119469	12/2016	Shimizu et al.	N/A	N/A
2017/0119470	12/2016	Diamant et al.	N/A	N/A
2017/0135709	12/2016	Nguyen et al.	N/A	N/A
2017/0151421	12/2016	Asher	N/A	N/A
2017/0192242	12/2016	Laycock	N/A	N/A
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2017/0303946	12/2016	Ku et al.	N/A	N/A

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2018/0008348	12/2017	Grace et al.	N/A	N/A
2018/0042661	12/2017	Long	N/A	N/A
2018/0042677	12/2017	Yu et al.	N/A	N/A
2018/0045897	12/2017	Chia	N/A	N/A
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2018/0085174	12/2017	Radtke et al.	N/A	N/A
2018/0092763	12/2017	Dagan et al.	N/A	N/A
2018/0095287	12/2017	Jeng et al.	N/A	N/A
2018/0098779	12/2017	Betelia et al.	N/A	N/A
2018/0152568	12/2017	Thumpudi et al.	N/A	N/A
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2019/0072378	12/2018	Hane et al.	N/A	N/A
2019/0097380	12/2018	Luft et al.	N/A	N/A
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2019/0321100	12/2018	Masotti et al.	N/A	N/A
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2020/0046429	12/2019	Tschida et al.	N/A	N/A
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2020/0129195	12/2019	McGowan et al.	N/A	N/A
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2022/0183738	12/2021	Flores et al.	N/A	N/A
2022/0218402	12/2021	Schultheis	N/A	N/A
2022/0249165	12/2021	Cook	N/A	N/A
2022/0249166	12/2021	Cook et al.	N/A	N/A
2022/0273324	12/2021	Schultheis	N/A	N/A
2022/0287732	12/2021	Anderson et al.	N/A	N/A
2022/0313293	12/2021	Singh	N/A	N/A
2022/0313359	12/2021	Schultheis et al.	N/A	N/A
2022/0338890	12/2021	Anderson et al.	N/A	N/A
2022/0354578	12/2021	Cook	N/A	N/A
2022/0387106	12/2021	Cook	N/A	N/A
2023/0013920	12/2022	Massimini	N/A	N/A
2023/0064371	12/2022	Cook et al.	N/A	N/A
2023/0137107	12/2022	Cook et al.	N/A	N/A
2023/0157754	12/2022	Bacher et al.	N/A	N/A

2023/0200906	12/2022	Cook et al.	N/A	N/A
2023/0233256	12/2022	Cook et al.	N/A	N/A
2023/0240748	12/2022	Cook et al.	N/A	N/A
2023/0248376	12/2022	Anderson et al.	N/A	N/A
2023/0255635	12/2022	Schultheis et al.	N/A	N/A
2023/0255688	12/2022	Schultheis et al.	N/A	N/A
2023/0255689	12/2022	Schultheis et al.	N/A	N/A
2023/0310054	12/2022	Schultheis	N/A	N/A
2023/0310067	12/2022	Schultheis et al.	N/A	N/A
2023/0310073	12/2022	Adams et al.	N/A	N/A
2023/0320576	12/2022	Feldman	N/A	N/A
2023/0338088	12/2022	Massimini et al.	N/A	N/A
2023/0338089	12/2022	Schultheis	N/A	N/A
2023/0414234	12/2022	Anderson et al.	N/A	N/A
2024/0001076	12/2023	Gelsinger	N/A	N/A
2024/0016544	12/2023	Schultheis et al.	N/A	N/A
2024/0016545	12/2023	Schultheis et al.	N/A	N/A
2024/0023813	12/2023	Milner	N/A	N/A
2024/0032995	12/2023	Schultheis et al.	N/A	N/A
2024/0033002	12/2023	Cook	N/A	N/A
2024/0041520	12/2023	Schultheis et al.	N/A	N/A
2024/0050170	12/2023	Fournier	N/A	N/A
2024/0050696	12/2023	Japuntich	N/A	N/A
2024/0058060	12/2023	Cook et al.	N/A	N/A
2024/0065712	12/2023	Schultheis	N/A	N/A
2024/0122648	12/2023	Cook	N/A	N/A
2024/0165658	12/2023	Fu	N/A	N/A
2024/0173526	12/2023	Kofidis	N/A	N/A
2024/0189543	12/2023	Salinas	N/A	N/A
2024/0216062	12/2023	Cook	N/A	N/A
2024/0260981	12/2023	Betelia	N/A	N/A
2024/0277410	12/2023	Cook	N/A	N/A
2024/0277974	12/2023	Oehler	N/A	N/A
2024/0277980	12/2023	O'Neill	N/A	N/A
2024/0285296	12/2023	Vo	N/A	N/A
2024/0285922	12/2023	Chu	N/A	N/A
2024/0325045	12/2023	Otake	N/A	N/A
2024/0382258	12/2023	Schultheis	N/A	N/A
2025/0025237	12/2024	Cook	N/A	N/A
2025/0040947	12/2024	Schultheis	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2017205323	12/2016	AU	N/A
2022227829	12/2021	AU	N/A
2229806	12/1996	CA	N/A
2281519	12/1997	CA	N/A
2983655	12/2015	CA	N/A
3209797	12/2021	CA	N/A
102057422	12/2010	CN	N/A

109223100	12/2018	CN	N/A
110638501	12/2019	CN	N/A
110638501	12/2019	CN	N/A
106794043	12/2019	CN	N/A
107411805	12/2021	CN	N/A
107899126	12/2021	CN	N/A
109475378	12/2021	CN	N/A
113876388	12/2021	CN	N/A
113877044	12/2021	CN	N/A
113907838	12/2021	CN	N/A
113951972	12/2021	CN	N/A
113951973	12/2021	CN	N/A
113974765	12/2021	CN	N/A
113974826	12/2021	CN	N/A
113993463	12/2021	CN	N/A
215384399	12/2021	CN	N/A
215386905	12/2021	CN	N/A
215458400	12/2021	CN	N/A
215458401	12/2021	CN	N/A
215505065	12/2021	CN	N/A
215534803	12/2021	CN	N/A
215537694	12/2021	CN	N/A
215584286	12/2021	CN	N/A
215606068	12/2021	CN	N/A
215651393	12/2021	CN	N/A
215651394	12/2021	CN	N/A
215651484	12/2021	CN	N/A
215653328	12/2021	CN	N/A
114053552	12/2021	CN	N/A
115175625	12/2021	CN	N/A
117752412	12/2023	CN	N/A
118055734	12/2023	CN	N/A
3038445	12/1981	DE	N/A
3836337	12/1989	DE	N/A
3913027	12/1989	DE	N/A
69431758	12/2002	DE	N/A
10230626	12/2003	DE	N/A
202008016760	12/2008	DE	N/A
102007046902	12/2008	DE	N/A
102008034702	12/2009	DE	N/A
102009007129	12/2009	DE	N/A
202010009899	12/2009	DE	N/A
102013201928	12/2013	DE	N/A
102020117713	12/2021	DE	N/A
0119296	12/1983	EP	N/A
0261831	12/1991	EP	N/A
558297	12/1992	EP	N/A
0571306	12/1992	EP	N/A
1179993	12/2001	EP	N/A
1946712	12/2007	EP	N/A

1946712	12/2007	EP	N/A
1453566	12/2007	EP	N/A
2157569	12/2009	EP	N/A
2879595	12/2014	EP	N/A
2879595	12/2014	EP	N/A
2944264	12/2014	EP	N/A
3226795	12/2016	EP	N/A
3266487	12/2017	EP	N/A
3318204	12/2017	EP	N/A
2879607	12/2018	EP	N/A
3461438	12/2018	EP	N/A
3473195	12/2018	EP	N/A
3240603	12/2018	EP	N/A
3643260	12/2019	EP	N/A
3240494	12/2020	EP	N/A
3076881	12/2021	EP	N/A
3932342	12/2021	EP	N/A
3936140	12/2021	EP	N/A
3960099	12/2021	EP	N/A
4051154	12/2021	EP	N/A
4129213	12/2022	EP	N/A
4277537	12/2022	EP	N/A
4297669	12/2023	EP	N/A
3182931	12/2023	EP	N/A
3950036	12/2023	EP	N/A
1082397	12/1966	GB	N/A
S62275446	12/1986	JP	N/A
05264763	12/1992	JP	N/A
H05264763	12/1992	JP	N/A
1996089511	12/1995	JP	N/A
H09117407	12/1996	JP	N/A
2004519296	12/2003	JP	N/A
2008506447	12/2007	JP	N/A
2008083273	12/2007	JP	N/A
2009519777	12/2008	JP	N/A
2009213589	12/2008	JP	N/A
2011524203	12/2010	JP	N/A
4805208	12/2010	JP	N/A
4808620	12/2010	JP	N/A
2014123147	12/2013	JP	N/A
A2014516614	12/2013	JP	N/A
A2015522344	12/2014	JP	N/A
2015217215	12/2014	JP	N/A
2018538077	12/2017	JP	N/A
2024511710	12/2023	JP	N/A
20050098932	12/2004	KR	N/A
20080040111	12/2007	KR	N/A
20160090877	12/2015	KR	N/A
20180054041	12/2017	KR	N/A
WO9007904	12/1989	WO	N/A

WO9105332	12/1990	WO	N/A
9203095	12/1991	WO	N/A
WO9208515	12/1991	WO	N/A
WO9524867	12/1994	WO	N/A
1999002095	12/1998	WO	N/A
1999020189	12/1998	WO	N/A
WO200067648	12/1999	WO	N/A
WO2000067648	12/1999	WO	N/A
WO0103599	12/2000	WO	N/A
WO0103599	12/2000	WO	N/A
20060006169	12/2005	WO	N/A
WO2006006169	12/2005	WO	N/A
WO2009121017	12/2008	WO	N/A
WO2009149321	12/2008	WO	N/A
WO2009152352	12/2008	WO	N/A
2010042653	12/2009	WO	N/A
WO2011094379	12/2010	WO	N/A
20110126580	12/2010	WO	N/A
WO2011126580	12/2010	WO	N/A
WO2012025833	12/2011	WO	N/A
WO2012042619	12/2011	WO	N/A
WO20120052924	12/2011	WO	N/A
WO2012058156	12/2011	WO	N/A
WO20120120495	12/2011	WO	N/A
WO2013119662	12/2012	WO	N/A
20130169807	12/2012	WO	N/A
WO2013169807	12/2012	WO	N/A
WO2014025397	12/2013	WO	N/A
WO20140022867	12/2013	WO	N/A
WO2014138582	12/2013	WO	N/A
WO2015056662	12/2014	WO	N/A
WO2015097251	12/2014	WO	N/A
20150177790	12/2014	WO	N/A
WO2016014999	12/2015	WO	N/A
WO2016089683	12/2015	WO	N/A
WO2016090175	12/2015	WO	N/A
WO2016098670	12/2015	WO	N/A
WO2016109739	12/2015	WO	N/A
WO2016143556	12/2015	WO	N/A
WO2016151595	12/2015	WO	N/A
WO20170192869	12/2016	WO	N/A
20180022641	12/2017	WO	N/A
WO2018022593	12/2017	WO	N/A
WO2018083666	12/2017	WO	N/A
20180175322	12/2017	WO	N/A
WO2018175322	12/2017	WO	N/A
WO2018191013	12/2017	WO	N/A
WO2019200201	12/2018	WO	N/A
WO2019222843	12/2018	WO	N/A
WO2020056031	12/2019	WO	N/A

WO20200086361	12/2019	WO	N/A
WO2020089876	12/2019	WO	N/A
WO2020157648	12/2019	WO	N/A
WO2020256693	12/2019	WO	N/A
WO2020256898	12/2019	WO	N/A
WO2020256898	12/2019	WO	N/A
WO2020256949	12/2019	WO	N/A
WO2020256949	12/2019	WO	N/A
WO2020263469	12/2019	WO	N/A
WO2020263685	12/2019	WO	N/A
WO2020263687	12/2019	WO	N/A
WO2020263688	12/2019	WO	N/A
WO2020263689	12/2019	WO	N/A
WO2021061451	12/2020	WO	N/A
WO2021067563	12/2020	WO	N/A
WO2021086571	12/2020	WO	N/A
WO2021096922	12/2020	WO	N/A
WO2021101766	12/2020	WO	N/A
WO2021101766	12/2020	WO	N/A
WO2021126762	12/2020	WO	N/A
WO2021150502	12/2020	WO	N/A
WO2021162855	12/2020	WO	N/A
WO2021173417	12/2020	WO	N/A
WO2021183367	12/2020	WO	N/A
WO2021183401	12/2020	WO	N/A
WO2021188233	12/2020	WO	N/A
WO2021231178	12/2020	WO	N/A
WO2021247685	12/2020	WO	N/A
WO2021257425	12/2020	WO	N/A
WO2022007490	12/2021	WO	N/A
WO2022008440	12/2021	WO	N/A
WO2022010767	12/2021	WO	N/A
WO2022055784	12/2021	WO	N/A
WO2022125525	12/2021	WO	N/A
WO2022154954	12/2021	WO	N/A
WO2022173719	12/2021	WO	N/A
WO2022183075	12/2021	WO	N/A
WO2022187058	12/2021	WO	N/A
WO2022216488	12/2021	WO	N/A
WO2022240674	12/2021	WO	N/A
WO2022260932	12/2021	WO	N/A
WO2023107334	12/2022	WO	N/A
WO2024079108	12/2023	WO	N/A

OTHER PUBLICATIONS

Machine Translation of JP05264763. (Year: 1993). cited by examiner

Stelzle, F., et al. “Diffuse Reflectance Spectroscopy for Optical Soft Tissue Differentiation as Remote Feedback Control for Tissue-Specific Laser Surgery”, Lasers in Surgery and Medicine, 2010, pp. 319-325, vol. 42, Wiley-Liss Inc. cited by applicant

Stelzle, F., et al. Tissue Discrimination by Uncorrected Autofluorescence Spectra: A Proof-of-

Principle Study for Tissue-Specific Laser Surgery, Sensors, 2013, pp. 13717-13731, vol. 13, Basel, Switzerland. cited by applicant

Tagawa, Y., et al. "Structure of laser-induced shock wave in water", Japan Society for the Promotion of Science, 2016. cited by applicant

Shen, Y., et al. "Theoretical and experimental studies of directivity of sound field generated by pulsed laser induced breakdown in liquid water", SPIE, 2013, pp. 8796141-8796148, vol. 8796, SPIE. cited by applicant

Preisack, M., et al. "Ultrafast imaging of tissue ablation by a XeCl excimer laser in saline", Lasers in Surgery and Medicine, 1992, pp. 520-527, vol. 12, Wiley-Liss Inc. cited by applicant

Versluis, M., et al. "How Snapping Shrimp Snap: Through Cavitating Bubbles", Science Mag, 2000, pp. 2114-2117, vol. 289, American Association for the Advancement of Science, Washington DC, USA. cited by applicant

Yan, D., et al. "Study of the Electrical Characteristics, Shock-Wave Pressure Characteristics, and Attenuation Law Based on Pulse Discharge in Water", Shock and Vibration, 2016, pp. 1-11, vol. 2016, Article ID 6412309, Hindawi Publishing Corporation. cited by applicant

Zhang, Q., et al. "Improved Instruments and Methods for the Photographic Study of Spark-Induced Cavitation Bubbles", Water, 2018, pp. 1-12, vol. 10, No. 1683. cited by applicant

"Damage threshold of fiber facets", NKT Photonics, 2012, pp. 1-4, Denmark. cited by applicant

Smith, A., et al. "Bulk and surface laser damage of silica by picosecond and nanosecond pulses at 1064 nm", Applied Optics, 2008, pp. 4812-4832, vol. 47, No. 26, Optical Society of America. cited by applicant

Smith, A., et al. "Deterministic Nanosecond Laser-Induced Breakdown Thresholds In Pure and Yb3 Doped Fused Silica", SPIE, 2007, pp. 6453171-64531712, vol. 6453, SPIE. cited by applicant

Sun, X., et al. "Laser Induced Damage to Large Core Optical Fiber by High Peak Power Laser", Specialty Photonics Division, 2010. cited by applicant

Smith, A., et al. "Nanosecond laser-induced breakdown in pure and Yb3 doped fused silica", SPIE, 2007, vol. 6403, SPIE. cited by applicant

Smith, A., et al. "Optical Damage Limits to Pulse Energy From Fibers", IEEE Journal of Selected Topics in Quantum Electronics, 2009, pp. 153-158, vol. 15, No. 1, IEEE. cited by applicant

Reichel, E., et al. "A Special Irrigation Liquid to Increase the Reliability of Laser-Induced Shockwave Lithotripsy", Lasers in Surgery and Medicine, 1992, pp. 204-209, vol. 12, Wiley-Liss Inc., Graz, Austria. cited by applicant

Reichel, E., et al. "Bifunctional irrigation liquid as an ideal energy converter for laser lithotripsy with nanosecond laser pulses", SPIE Lasers in Urology, Laparoscopy, and General Surgery, 1991, pp. 129-133, vol. 1421, SPIE. cited by applicant

Reichel, E., et al. "Laser-induced Shock Wave Lithotripsy with a Regenerative Energy Converter", Lasers in Medical Science, 1992, pp. 423-425, vol. 7, Bailliere Tindall. cited by applicant

Hardy, L., et al. "Cavitation Bubble Dynamics during Thulium Fiber Laser Lithotripsy", SPIE BiOS, 2016, vol. 9689, SPIE. cited by applicant

Deckelbaum, L., "Coronary Laser Angioplasty", Lasers in Surgery and Medicine, 1994, pp. 101-110, vol. 14, Wiley-Liss Inc., Conneticut, USA. cited by applicant

Shangguan, H., et al. "Effects of Material Properties on Laser-induced Bubble Formation in Absorbing Liquids and On Submerged Targets", Diagnostic and Therapeutic Cardiovascular Interventions VII, SPIE, 1997, pp. 783-791, vol. 2869, SPIE. cited by applicant

Van Leeuwen, T., et al. "Excimer Laser Induced Bubble: Dimensions, Theory, and Implications for Laser Angioplasty", Lasers in Surgery and Medicine, 1996, pp. 381-390, vol. 18, Wiley-Liss Inc., The Netherlands. cited by applicant

Vogel, A., et al. "Minimization of Cavitation Effects in Pulsed Laser Ablation Illustrated on Laser Angioplasty", Applied Physics, 1996, pp. 173-182, vol. 62, Springer-Verlag. cited by applicant

Vogel, A., et al. "Shock Wave Emission and Cavitation Bubble Generation by Picosecond and

Nanosecond Optical Breakdown in Water”, The Journal of Acoustical Society of America, 1996, pp. 148-165, vol. 100, No. 1, The Acoustical Society of America. cited by applicant

Varghese, B., et al. “Influence of absorption induced thermal initiation pathway on irradiance threshold for laser induced breakdown”, Biomedical Optics Express, 2015, vol. 6, No. 4, Optical Society of America. cited by applicant

Linz, N., et al. “Wavelength dependence of nanosecond infrared laser-induced breakdown in water: Evidence for multiphoton initiation via an intermediate state”, Physical Review, 2015, pp. 134114.1-134114.10, vol. 91, American Physical Society. cited by applicant

International Search Report and Written Opinion dated Jun. 27, 2018, in PCT Application Serial No. PCT/US2018/027121. cited by applicant

International Search Report and Written Opinion dated Jul. 20, 2018, in PCT Application Serial No. PCT/US2018/027801. cited by applicant

International Search Report and Written Opinion dated Jul. 20, 2018, in PCT Application Serial No. PCT/US2018/027784. cited by applicant

European Search Report, for European Patent Application No. 18185152, mailed Dec. 13, 2018. cited by applicant

International Search Report and Written Opinion dated May 22, 2019, in PCT Application Serial No. PCT/US2019/022009. cited by applicant

International Search Report and Written Opinion dated May 29, 2019, in PCT Application Serial No. PCT/US2019/022016. cited by applicant

International Search Report and Written Opinion dated Jun. 22, 2018, in Application Serial No. NL2019807, issued by the European Patent Office. cited by applicant

Noimark, Sacha, et al., “Carbon-Nanotube-PDMS Composite Coatings on Optical Fibers for All-Optical Ultrasound Imaging”, Advanced Functional Materials, 2016, pp. 8390-8396, vol. 26, Wiley-Liss Inc. cited by applicant

Chen, Sung-Liang, “Review of Laser-Generated Ultrasound Transmitters and their Applications to All-Optical Ultrasound Transducers and Imaging”, Appl. Sci. 2017, 7, 25. cited by applicant

Colchester, R., et al. “Laser-Generated ultrasound with optica fibres using functionalised carbon nanotube composite coatings”, Appl. Phys. Lett., 2014, vol. 104, 173504, American Institute of Physics. cited by applicant

Poduval, R., et al. “Optical fiber ultrasound transmitter with electrospun carbon nanotube-polymer composite”, Appl. Phys. Lett., 2017, vol. 110, 223701, American Institute of Physics. cited by applicant

Tian, J., et al. “Distributed fiber-optic laser-ultrasound generation based on ghost-mode of tilted fiber Bragg gratings”, Optics Express, Mar. 2013, pp. 6109-6114, vol. 21, No. 5, Optical Society of America. cited by applicant

Kim, J., et al. “Optical Fiber Laser-Generated-Focused-Ultrasound Transducers for Intravascular Therapies”, IEEE, 2017. cited by applicant

Kang, H., et al. “Enhanced photocoagulation with catheter-based diffusing optical device”, Journal of Biomedical Optics, 2012, vol. 17, Issue 11, 118001, SPIE. cited by applicant

International Search Report and Written Opinion dated Jan. 3, 2020, in PCT Application Serial No. PCT/US2019/056579. cited by applicant

Communication Pursuant to Article 94(3) EPC, for European Patent Application No. 18185152.8, mailed Jan. 16, 2019. cited by applicant

European Search Report, for European Patent Application No. 18185152.8, mailed Dec. 20, 2018. cited by applicant

International Search Report and Written Opinion dated Jul. 29, 2020 in PCT Application Serial No. PCT/US2020/034005. cited by applicant

International Search Report and Written Opinion dated Sep. 11, 2020 in PCT Application Serial No. PCT/US2020/038517. cited by applicant

International Search Report and Written Opinion dated Sep. 9, 2020 in PCT Application Serial No. PCT/US2020/038530. cited by applicant

International Search Report and Written Opinion dated Sep. 11, 2020 in PCT Application Serial No. PCT/US2020/038521. cited by applicant

International Search Report and Written Opinion dated Sep. 7, 2020 in PCT Application Serial No. PCT/US2020/034642. cited by applicant

Vogel, A., et al. "Intraocular Photodisruption With Picosecond and Nanosecond Laser Pulses: Tissue Effects in Cornea, Lens, and Retina", *Investigative Ophthalmology & Visual Science*, Jun. 1994, pp. 3032-3044, vol. 35, No. 7, Association for Research in Vision and Ophthalmology. cited by applicant

Jones, H. M., et al. "Pulsed dielectric breakdown of pressurized water and salt solutions", *Journal of Applied Physics*, Jun. 1998, pp. 795-805, vol. 77, No. 2, American Institute of Physics. cited by applicant

Kozulin, I., et al. "The dynamic of the water explosive vaporization on the flat microheater", *Journal of Physics: Conference Series*, 2018, pp. 1-4, IOP Publishing, Russia. cited by applicant

Cross, F., "Laser Angioplasty", *Vascular Medicine Review*, 1992, pp. 21-30, Edward Arnold. cited by applicant

Doukas, A. G., et al. "Laser-generated stress waves and their effects on the cell membrane", *IEEE Journal of Selected Topics in Quantum Electronics*, 1999, pp. 997-1003, vol. 5, Issue 4, IEEE. cited by applicant

Noack, J., et al. "Laser-Induced Plasma Formation in Water at Nanosecond to Femtosecond Time Scales: Calculation of Thresholds, Absorption Coefficients, and Energy Density", *IEEE Journal of Quantum Electronics*, 1999, pp. 1156-1167, vol. 35, No. 8, IEEE. cited by applicant

Pratsos, A., "The use of Laser for the treatment of coronary artery disease", Bryn Mawr Hospital, 2010. cited by applicant

Li, Xian-Dong, et al. "Influence of deposited energy on shock wave induced by underwater pulsed current discharge", *Physics of Plasmas*, 2016, vol. 23, American Institute of Physics. cited by applicant

Logunov, S., et al. "Light diffusing optical fiber illumination", *Renewable Energy and the Environment Congress*, 2013, Corning, NY, USA. cited by applicant

Maxwell, A. D., et al. "Cavitation clouds created by shock scattering from bubbles during histotripsy", *Acoustical Society of America*, 2011, pp. 1888-1898, vol. 130, No. 4, Acoustical Society of America. cited by applicant

McAteer, James A., et al. "Ultracal-30 Gypsum Artificial Stones For Research On The Mechanisms Of Stone Breakage In Shock Wave Lithotripsy", 2005, pp. 429-434, Springer-Verlag. cited by applicant

Vogel, A., et al. "Mechanisms of Intraocular Photodisruption With Picosecond and Nanosecond Laser Pulses", *Lasers in Surgery and Medicine*, 1994, pp. 32-43, vol. 15, Wiley-Liss Inc., Lubeck, Germany. cited by applicant

Vogel, A., et al. "Mechanisms of Pulsed Laser Ablation of Biological Tissues", *Chemical Reviews*, 2003, pp. 577-644, vol. 103, No. 2, American Chemical Society. cited by applicant

Medlight, "Cylindrical light diffuser Model RD-ML", Medlight S.A., Switzerland. 2015. cited by applicant

Medlight, "Cylindrical light diffuser Model RD", Medlight S.A., Switzerland. 2015. cited by applicant

Mayo, Michael E., "Interaction of Laser Radiation with Urinary Calculi", *Cranfield University Defense and Security, PhD Thesis*, 2009, Cranfield University. cited by applicant

Mirshekari, G., et al. "Microscale Shock Tube", *Journal of Microelectromechanical Systems*, 2012, pp. 739-747, vol. 21, No. 3, IEEE. cited by applicant

"Polymicro Sculpted Silica Fiber Tips", Molex, 2013, Molex. cited by applicant

Zhou, J., et al. "Optical Fiber Tips and Their Applications", Polymicro Technologies A Subsidiary of Molex, Nov. 2007. cited by applicant

Liang, Xiao-Xuan, et al. "Multi-Rate-Equation modeling of the energy spectrum of laser-induced conduction band electrons in water", Optics Express, 2019, vol. 27, No. 4, Optical Society of America. cited by applicant

Nachabe, R., et al. "Diagnosis of breast cancer using diffuse optical spectroscopy from 500 to 1600 nm: comparison of classification methods", Journal of Biomedical Optics, 2011, vol. 16(8), SPIE. cited by applicant

Naugol'Nykh, K. A., et al. "Spark Discharges in Water", Academy of Sciences USSR Institute of Acoustics, 1971, Nauka Publishing Co., Moscow, USSR. cited by applicant

Van Leeuwen, Ton G., et al. "Noncontact Tissue Ablation by Holmium: YSGG Laser Pulses in Blood", Lasers in Surgery and Medicine, 1991, vol. 11, pp. 26-34, Wiley-Liss Inc. cited by applicant

Nyame, Yaw A., et al. "Kidney Stone Models for In Vitro Lithotripsy Research: A Comprehensive Review", Journal of Endourology, Oct. 2015, pp. 1106-1109, vol. 29, No. 10, Mary Ann Liebert Inc., Cleveland, USA. cited by applicant

Ohl, Siew-Wan, et al. "Bubbles with shock waves and ultrasound: a review", Interface Focus, pp. 1-15, vol. 5, The Royal Society Publishing. Oct. 2015. cited by applicant

Zheng, W., "Optical Lenses Manufactured on Fiber Ends", IEEE, 2015, Splicer Engineering, Duncan SC USA. cited by applicant

Dwyer, P. J., et al. "Optically integrating balloon device for photodynamic therapy", Lasers in Surgery and Medicine, 2000, pp. 58-66, vol. 26, Issue 1, Wiley-Liss Inc., Boston MA USA. cited by applicant

"The New Optiguide DCYL700 Fiber Optic Diffuser Series", Optiguide Fiber Optic Spec Sheet, Pinnacle Biologics, 2014, Pinnacle Biologics, Illinois, USA. cited by applicant

Van Leeuwen, Ton G., et al. "Origin of arterial wall dissections induced by pulsed excimer and mid-infrared laser ablation in the pig", JACC, 1992, pp. 1610-1618, vol. 19, No. 7, American College of Cardiology. cited by applicant

Oshita, D., et al. "Characteristic of Cavitation Bubbles and Shock Waves Generated by Pulsed Electric Discharges with Different Voltages", IEEE, 2012, pp. 102-105, Kumamoto, Japan. cited by applicant

Karsch, Karl R., et al. "Percutaneous Coronary Excimer Laser Angioplasty in Patients With Stable and Unstable Angina Pectoris", Circulation, 1990, pp. 1849-1859, vol. 81, No. 6, American Heart Association, Dallas TX, USA. cited by applicant

Murray, A., et al. "Peripheral laser angioplasty with pulsed dye laser and ball tipped optical fibres", The Lancet, 1989, pp. 1471-1474, vol. 2, Issue 8678-8679. cited by applicant

Mohammadzadeh, M., et al. "Photoacoustic Shock Wave Emission and Cavitation from Structured Optical Fiber Tips", Applied Physics Letters, 2016, vol. 108, American Institute of Physics Publishing LLC. cited by applicant

Doukas, A. G., et al. "Physical characteristics and biological effects of laser-induced stress waves", Ultrasound in Medicine and Biology, 1996, pp. 151-164, vol. 22, Issue 2, World Federation for Ultrasound in Medicine and Biology, USA. cited by applicant

Doukas, A. G., et al. "Physical factors involved in stress-wave-induced cell injury: the effect of stress gradient", Ultrasound in Medicine and Biology, 1995, pp. 961-967, vol. 21, Issue 7, Elsevier Science Ltd., USA. cited by applicant

Piedrahita, Francisco S., "Experimental Research Work On A Sub-Millimeter Spark-Gap For Sub Nanosecond Gas Breakdown", Thesis for Universidad Nacional De Colombia, 2012, Bogota, Colombia. cited by applicant

Vogel, A., et al. "Plasma Formation in Water by Picosecond and Nanosecond Nd: YAG Laser Pulses—Part I: Optical Breakdown at Threshold and Superthreshold Irradiance", IEEE Journal of

Selected Topics in Quantum Electronics, 1996, pp. 847-859, vol. 2, No. 4, IEEE. cited by applicant

Park, Hee K., et al. "Pressure Generation and Measurement in the Rapid Vaporization of Water on a Pulsed-Laser-Heated Surface", *Journal of Applied Physics*, 1996, pp. 4072-4081, vol. 80, No. 7, American Institute of Physics. cited by applicant

Cummings, Joseph P., et al. "Q-Switched laser ablation of tissue: plume dynamics and the effect of tissue mechanical properties", *SPIE, Laser-Tissue Interaction III*, 1992, pp. 242-253, vol. 1646. cited by applicant

Lee, Seung H., et al. "Radial-firing optical fiber tip containing conical-shaped air-pocket for biomedical applications", *Optics Express*, 2015, vol. 23, No. 16, Optical Society of America. cited by applicant

Hui, C., et al. "Research on sound fields generated by laser-induced liquid breakdown", *Optica Applicata*, 2010, pp. 898-907, vol. XL, No. 4, Xi'an, China. cited by applicant

Riel, Louis-Philippe, et al. "Characterization of Calcified Plaques Retrieved From Occluded Arteries and Comparison with Potential Artificial Analogues", *Proceedings of the ASME 2014 International Mechanical Engineering Congress and Exposition*, 2014, pp. 1-11, ASME, Canada. cited by applicant

Roberts, Randy M., et al. "The Energy Partition of Underwater Sparks", *The Journal of the Acoustical Society of America*, 1996, pp. 3465-3475, vol. 99, No. 6, Acoustical Society of America. cited by applicant

Rocha, R., et al. "Fluorescence and Reflectance Spectroscopy for Identification of Atherosclerosis in Human Carotid Arteries Using Principal Components Analysis", *Photomedicine and Laser Surgery*, 2008, pp. 329-335, vol. 26, No. 4, Mary Ann Liebert Inc. cited by applicant

Scepanovic, Obrad R., et al. "Multimodal spectroscopy detects features of vulnerable atherosclerotic plaque", *Journal of Biomedical Optics*, 2011, pp. 1-10, vol. 16, No. 1, SPIE. cited by applicant

Serruys, P. W., et al. "Shaking and Breaking Calcified Plaque Lithoplasty, a Breakthrough in Interventional Armamentarium?", *JACC: Cardiovascular Imaging*, 2017, pp. 907-911, vol. 10, No. 8, Elsevier. cited by applicant

Vogel, A., et al. "Shock wave emission and cavitation bubble generation by picosecond and nanosecond optical breakdown in water", *The Journal of the Acoustical Society of America*, 1996, pp. 148-165, vol. 100, No. 1, Acoustical Society of America. cited by applicant

Vogel, A., et al. "Shock-Wave Energy and Acoustic Energy Dissipation After Laser-induced Breakdown", *SPIE*, 1998, pp. 180-189, vol. 3254, SPIE. cited by applicant

International Search Report and Written Opinion, issued by the EP/ISA, in PCT/US2021/048819, dated Jan. 14, 2022. cited by applicant

Shen, Yajie et al. "High-peak-power and narrow-linewidth Q-switched Ho: YAG laser in-band pumped at 1931 nm." *Applied Physics Express* 13.5 (2020): 052006. (Year 2020). cited by applicant

Davletshin, Yevgeniy R., "A Computational Analysis of Nanoparticle-Mediated Optical Breakdown", A dissertation presented to Ryerson University in Partial Fulfillment of the requirements for the degree of Doctor of Philosophy in the Program of Physics, Toronto, Ontario, CA 2017. cited by applicant

Vogel, A., et al. "Acoustic transient generation by laser-produced cavitation bubbles near solid boundaries", *Journal Acoustical Society of America*, 1988, pp. 719-731, vol. 84. cited by applicant

Asshauer, T., et al. "Acoustic transient generation by holmium-laser-induced cavitation bubbles", *Journal of Applied Physics*, Nov. 1, 1994, pp. 5007-5013, vol. 76, No. 9, American Institute of Physics. cited by applicant

Zheng, W., "Optic Lenses Manufactured on Fiber Ends", 2015, Splicer Engineering AFL, Duncan, SC USA. cited by applicant

Ali, Ziad A., et al. "Optical Coherence Tomography Characterization of Coronary Lithoplasty for

Treatment of Calcified Lesions”, JACC: Cardiovascular Imaging, 2017, pp. 897-906, vol. 109, No. 8, Elsevier. cited by applicant

Ali, Ziad A., et al. “Intravascular lithotripsy for treatment of stent underexpansion secondary to severe coronary calcification” 2018, European Society of Cardiology. cited by applicant

Ashok, Praveen C., et al. “Raman spectroscopy bio-sensor for tissue discrimination in surgical robotics—full article”, Journal of Biophotonics, 2014, pp. 103-109, vol. 7, No. 1-2. cited by applicant

Ashok, Praveen C., et al. “Raman spectroscopy bio-sensor for tissue discrimination in surgical robotics—proof” Journal of Biophotonics 7, 2014, No. 1-2. cited by applicant

Bian, D. C., et al. “Experimental Study of Pulsed Discharge Underwater Shock-Related Properties in Pressurized Liquid Water”, Hindawi Advances in Materials Science and Engineering, Jan. 2018, 12 pages, vol. 2018, Article ID 8025708. cited by applicant

Bian, D. C., et al. “Study on Breakdown Delay Characteristics Based on High-voltage Pulse Discharge in Water with Hydrostatic Pressure”, Journal of Power Technologies 97(2), 2017, pp. 89-102. cited by applicant

Doukas, A. G., et al. “Biological effects of laser induced shock waves: Structural and functional cell damage in vitro”, Ultrasound in Medicine and Biology, 1993, pp. 137-146, vol. 19, Issue 2, Pergamon Press, USA. cited by applicant

Brodmann, Marianne et al. “Safety and Performance of Lithoplasty for Treatment of Calcified Peripheral Artery Lesions”, JACC, 2017, vol. 70, No. 7. cited by applicant

Brouillette, M., “Shock Waves at Microscales”, 2003, pp. 3-12, Springer-Verlag. cited by applicant

Mirshekari, G., et al. “Shock Waves in Microchannels”, 2013, pp. 259-283, vol. 724, Cambridge University Press. cited by applicant

“Bubble Dynamics and Shock Waves”, Springer, 2013, Springer-Verlag, Berlin Heidelberg. cited by applicant

Hardy, Luke A., et al. “Cavitation Bubble Dynamics During Thulium Fiber Laser Lithotripsy”, SPIE, Feb. 29, 2016, vol. 9689, San Francisco, California, USA. cited by applicant

Claverie, A., et al. “Experimental characterization of plasma formation and shockwave propagation induced by high power pulsed underwater electrical discharge”, Review of Scientific Instruments, 2014, American Institute of Physics. cited by applicant

Blackmon, Richard L., et al. “Comparison of holmium: YAG and thulium fiber laser lithotripsy ablation thresholds, ablation rates, and retropulsion effects”, Journal of Biomedical Optics, 2011, vol. 16(7), SPIE. cited by applicant

Debasis, P., et al. “Continuous-wave and quasi-continuous wave thulium-doped all-fiber laser: implementation on kidney stone fragmentations”, Applied Optics, Aug. 10, 2016, vol. 55, No. 23, Optical Society of America. cited by applicant

Cook, Jason R., et al. “Tissue mimicking phantoms for photoacoustic and ultrasonic imaging”, Biomedical Optics Express, 2011, vol. 2, No. 11, Optical Society of America. cited by applicant

Deckelbaum, Lawrence I., “Coronary Laser Angioplasty”, Lasers in Surgery and Medicine, 1994, pp. 101-110, Wiley-Liss Inc. cited by applicant

Costanzo, F., “Underwater Explosion Phenomena and Shock Physics”, Research Gate, 2011. cited by applicant

Mizeret, J. C., et al. “Cylindrical fiber optic light diffuser for medical applications”, Lasers in Surgery and Medicine, 1996, pp. 159-167, vol. 19, Issue 2, Wiley-Liss Inc., Lausanne, Switzerland. cited by applicant

De Silva, K., et al. “A Calcific, Undilatable Stenosis Lithoplasty, a New Tool in the Box?”, JACC: Cardiovascular Interventions, 2017, vol. 10, No. 3, Elsevier. cited by applicant

Vesselov, L., et al. “Design and performance of thin cylindrical diffusers created in Ge-doped multimode optical fibers”, Applied Optics, 2005, pp. 2754-2758, vol. 44, Issue 14, Optical Society of America. cited by applicant

Hutchens, Thomas C., et al. "Detachable fiber optic tips for use in thulium fiber laser lithotripsy", *Journal of Biomedical Optics*, Mar. 2013, vol. 18(3), SPIE. cited by applicant

Kostanski, Kris L., et al. "Development of Novel Tunable Light Scattering Coating Materials for Fiber Optic Diffusers in Photodynamic Cancer Therapy", *Journal of Applied Polymer Science*, 2009, pp. 1516-1523, vol. 112, Wiley InterScience. cited by applicant

Kristiansen, M., et al. "High Voltage Water Breakdown Studies", DoD, 1998, Alexandria, VA, USA. cited by applicant

Dwyer, J. R., et al. "A study of X-ray emission from laboratory sparks in air at atmospheric pressure", *Journal of Geophysical Research*, 2008, vol. 113, American Geophysical Union. cited by applicant

Jansen, Duco E., et al. "Effect of Pulse Duration on Bubble Formation and Laser-Induced Pressure Waves During Holmium Laser Ablation", *Lasers in Surgery and Medicine* 18, 1996, pp. 278-293, Wiley-Liss Inc., Austin, TX, USA. cited by applicant

Shangguan, HanQun et al. "Effects of Material Properties on Laser-induced Bubble Formation in Absorbing Liquids and On Submerged Targets", SPIE, 1997, pp. 783-791, vol. 2869. cited by applicant

Varghese, B., et al. "Effects of polarization and absorption on laser induced optical breakdown threshold for skin rejuvenation", SPIE, Mar. 9, 2016, vol. 9740, SPIE, San Francisco, USA. cited by applicant

Varghese, B., et al. "Effects of polarization and apodization on laser induced optical breakdown threshold", *Optics Express*, Jul. 29, 2013, vol. 21, No. 15, Optical Society of America. cited by applicant

Bonito, Valentina, "Effects of polarization, plasma and thermal initiation pathway on irradiance threshold of laser induced optical breakdown", Philips Research, 2013, The Netherlands. cited by applicant

Vogel, A. et al. "Energy balance of optical breakdown in water at nanosecond to femtosecond time scales", *Applied Physics B* 68, 1999, pp. 271-280, Springer-Verlag. cited by applicant

Kang, Hyun W., et al. "Enhanced photocoagulation with catheter based diffusing optical device", *Journal of Biomedical Optics*, Nov. 2012, vol. 17(11), SPIE. cited by applicant

Esch, E., et al. "A Simple Method For Fabricating Artificial Kidney Stones Of Different Physical Properties", National Institute of Health Public Access Author Manuscript, Aug. 2010. cited by applicant

Isner, Jeffrey M., et al. "Excimer Laser Atherectomy", *Circulation*, Jun. 1990, vol. 81, No. 6, American Heart Association, Dallas, TX, USA. cited by applicant

Israel, Douglas H., et al. "Excimer Laser-Facilitated Balloon Angioplasty of a Nondilatable Lesion", *JACC*, Oct. 1991, vol. 18, No. 4, American College of Cardiology, New York, USA. cited by applicant

Van Leeuwen, Ton G., et al. "Excimer Laser Induced Bubble: Dimensions, Theory, and Implications for Laser Angioplasty", *Lasers in Surgery and Medicine* 18, 1996, pp. 381-390, Wiley-Liss Inc., Utrecht, The Netherlands. cited by applicant

Nguyen, H., et al. "Fabrication of multipoint side-firing optical fiber by laser micro-ablation", *Optics Letters*, May 1, 2017, vol. 42, No. 9, Optical Society of America. cited by applicant

Zheng, W., "Optic Lenses Manufactured on Fiber Ends", 2015, IEEE, Duncan, SC, USA. cited by applicant

Whitesides, George M., et al. "Fluidic Optics", 2006, vol. 6329, SPIE, Cambridge, MA, USA. cited by applicant

Forero, M., et al. "Coronary lithoplasty: a novel treatment for stent underexpansion", *Cardiovascular Flashlight*, 2018, European Society of Cardiology. cited by applicant

Ghanate, A. D., et al. "Comparative evaluation of spectroscopic models using different multivariate statistical tools in a multicancer scenario", *Journal of Biomedical Optics*, Feb. 2011, pp. 1-9, vol.

16(2), SPIE. cited by applicant

Roberts, Randy M., et al. "The Energy Partition of Underwater Sparks", The Journal of the Acoustical Society of America, Jun. 1996, pp. 3465-3474, Acoustical Society of America, Austin, TX, USA. cited by applicant

Blackmon, Richard L., et al. "Holmium: YAG Versus Thulium Fiber Laser Lithotripsy", Lasers in Surgery and Medicine, 2010, pp. 232-236, Wiley-Liss Inc. cited by applicant

Varghese, B., "Influence of absorption induced thermal initiation pathway on irradiance threshold for laser induced breakdown", Biomedical Optics Express, 2015, vol. 6, No. 4, Optical Society of America. cited by applicant

Noack, J., "Influence of pulse duration on mechanical effects after laser-induced breakdown in water", Journal of Applied Physics, 1998, pp. 7488-EOA, vol. 83, American Institute of Physics. cited by applicant

Van Leeuwen, Ton G., et al. "Intraluminal Vapor Bubble Induced by Excimer Laser Pulse Causes Microsecond Arterial Dilation and Invagination Leading to Extensive Wall Damage in the Rabbit", Circulation, Apr. 1993, vol. 87, No. 4, American Heart Association, Dallas, TX, USA. cited by applicant

PathFinder Digital, "Free Space Optics vs. Fiber Optics", 2023. cited by applicant

International Search Report and Written Opinion, issued in Application Serial No. PCT/US2023/016152, dated Jul. 12, 2023. cited by applicant

"Custom Medical Skived Tubing", Duke Extrusion, 2025. <https://www.dukeextrusion.com/tubing-options/skived-tubing>. cited by applicant

International Preliminary Report on Patentability dated Sep. 15, 2020 in PCT Application Serial No. PCT/US2019/022009. cited by applicant

International Search Report and Written Opinion dated Sep. 14, 2020 in PCT Application Serial No. PCT/US2020/038523. cited by applicant

International Search Report and Written Opinion dated Oct. 2, 2020 in PCT Application Serial No. PCT/US2020/036107. cited by applicant

Schafter+Kirchhoff, Laser Beam Couplers series 60SMS for coupling into single-mode and polarization-maintaining fiber cables, Schafter+Kirchhoff, pp. 1-5, Germany. Dec. 2, 2021. cited by applicant

International Search Report and Written Opinion dated Jan. 29, 2020 in PCT Application Serial No. PCT/US2020/059961. cited by applicant

International Search Report and Written Opinion dated Jan. 20, 2020 in PCT Application Serial No. PCT/US2020/054792. cited by applicant

Partial Search Report and Provisional Opinion dated Feb. 19, 2021 in PCT Application Serial No. PCT/US2020/059960. cited by applicant

Shariat, Mohammad H., et al. "Localization of the ectopic spiral electrical source using intracardiac electrograms during atrial fibrillation." 2015 IEEE 28th Canadian Conference on Electrical and Computer Engineering (CCECE). IEEE, 2015. cited by applicant

Nademanee, Koonlawee, et al. "A new approach for catheter ablation of atrial fibrillation: mapping of the electrophysiologic substrate." Journal of the American College of Cardiology 43.11 (2004): 2044-2053. cited by applicant

Calkins, Hugh. "Three dimensional mapping of atrial fibrillation: techniques and necessity." Journal of interventional cardiac electrophysiology 13.1 (2005): 53-59. cited by applicant

Shariat, Mohammad Hassan. Processing the intracardiac electrogram for atrial fibrillation ablation. Diss. Queen's University (Canada), 2016. cited by applicant

Meng et al., "Accurate Recovery Of Atrial Endocardial Potential Maps From Non-contact Electrode Data." Auckland Bioengineering Institute. (ID 1421). May 2019. cited by applicant

Jiang et al., "Multielectrode Catheter For Substrate Mapping For Scar-related VT Ablation: A Comparison Between Grid Versus Linear Configurations." UChicago Medicine, Center for

Arrhythmia Care, Chicago IL (ID 1368). Poster for conference in San Francisco, May 8-11, 2019. cited by applicant

Sacher et al., "Comparison Of Manual Vs Automatic Annotation To Identify Abnormal Substrate For Scar Related VT Ablation." LIRYC Institute, Bordeaux University Hospital, France (ID 1336). Poster for conference in San Francisco, May 8-11, 2019. cited by applicant

Oriel Instruments, "Introduction to Beam Splitters for Optical Research Applications", Apr. 2014, pp. 1-9, <https://www.azoptics.com/Article.aspx?ArticalID=871>. cited by applicant

International Search Report and Written Opinion dated Apr. 12, 2021 in PCT Application Serial No. PCT/US2020/059960. cited by applicant

International Search Report and Written Opinion dated Apr. 13, 2021 in PCT Application Serial No. PCT/US2020/064846. cited by applicant

International Search Report and Written Opinion dated Apr. 13, 2021 in PCT Application Serial No. PCT/US2021/013944. cited by applicant

International Search Report and Written Opinion dated May 25, 2021 in PCT Application Serial No. PCT/US2021/017604. cited by applicant

International Search Report and Written Opinion dated Jun. 2, 2021 in PCT Application Serial No. PCT/US2021/018522. cited by applicant

International Search Report and Written Opinion dated Jun. 2, 2021 in PCT Application Serial No. PCT/US2021/015204. cited by applicant

International Search Report and Written Opinion dated Jun. 17, 2021 in PCT Application Serial No. PCT/US2021/020934. cited by applicant

International Search Report and Written Opinion dated Jul. 13, 2021 in PCT Application Serial No. PCT/US2021/024216. cited by applicant

International Search Report and Written Opinion dated Jun. 22, 2021 in PCT Application Serial No. PCT/US2021/020937. cited by applicant

International Search Report and Written Opinion dated Jun. 24, 2021 in PCT Application Serial No. PCT/US2021/021272. cited by applicant

International Search Report and Written Opinion dated Aug. 20, 2021 in PCT Application Serial No. PCT/US2021/031130. cited by applicant

International Search Report and Written Opinion dated Apr. 4, 2022 in PCT Application Serial No. PCT/US2021/062170. cited by applicant

International Search Report and Written Opinion dated Apr. 4, 2022 in PCT Application Serial No. PCT/US2021/065073. cited by applicant

Partial Search Report and Provisional Opinion dated May 3, 2022 in PCT Application No. PCT/US2022/015577. cited by applicant

International Search Report and Written Opinion dated May 13, 2022 in PCT Application Serial No. PCT/US2022/017562. cited by applicant

International Search Report and Written Opinion dated Jun. 28, 2022, in PCT Application Serial No. PCT/US2022/015577. cited by applicant

International Search Report and Written Opinion dated Jun. 27, 2022, in PCT Application Serial No. PCT/US2022/022460. cited by applicant

International Search Report and Written Opinion dated Aug. 25, 2022 in PCT Application Serial No. PCT US/2022/028035. cited by applicant

International Search Report and Written Opinion dated Sep. 15, 2022 in PCT Application Serial No. PCT US/2022/032045. cited by applicant

International Search Report and Written Opinion dated Nov. 8, 2022 in PCT Application Serial No. PCT US/2022/039678. cited by applicant

International Search Report and Written Opinion, PCT Application Serial No. PCT/US2022/047751 issued Feb. 10, 2023, by the European Patent Office. cited by applicant

International Search Report and Written Opinion, PCT Application Serial No.

PCT/US2022/047691 issued Feb. 13, 2023, by the European Patent Office. cited by applicant
Accucoat, “Beamsplitter: Divide, combine & conquer”; 2023. cited by applicant
Lin et al., “Photoacoustic imaging”, Science Direct; 2021. cited by applicant
Zhou et al., “Photoacoustic Imaging with fiber optic technology: A review”, Science Direct; 2020. cited by applicant
International Search Report and Written Opinion issued by the European Patent Office, for Serial No. PCT/US2022/053775, dated Apr. 21, 2023. cited by applicant
International Search Report and Written Opinion issued by the European Patent Office, for Serial No. PCT/US2023/011497, dated Apr. 28, 2023. cited by applicant
International Search Report and Written Opinion issued by the European Patent Office, for Serial No. PCT/US2023/012599, dated May 19, 2023. cited by applicant
All Foreign References and Non-Patent Literature Are Available in the Parent Application, U.S. Appl. No. 16/874,065. cited by applicant
Definition of ablation—NCI Dictionary of Cancer Terms—NCI, National Cancer Institute, p. 1 (Year:2025). cited by applicant
Daemen, J., Tovar Forero, M.N, “The Coronary Intravascular Lithotripsy System”, ICR Journal, 2019; 14(3); 174-181. cited by applicant
Butt, N., Khalid, N., Shlofmitz, E., “Intravascular Lithotripsy”; NCBI Bookshelf. A service of the National Library of Medicine, National Institutes of Health; StatPearls Publishing, 2023. cited by applicant

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Background/Summary

RELATED APPLICATIONS (1) The present application is a continuation-in-part application claiming the benefit under 35 U.S.C. 120 on co-pending U.S. patent application Ser. No. 16/874,065, filed on May 13, 2020. Additionally, U.S. patent application Ser. No. 16/874,065 claims priority on U.S. Provisional Application Ser. No. 62/863,506, filed on Jun. 19, 2019, and on U.S. Provisional Application Ser. No. 62/965,069, filed on Jan. 23, 2020. To the extent permitted, the contents of U.S. patent application Ser. No. 16/874,065 and U.S. Provisional Application Ser. Nos. 62/863,506 and 62/965,069 are incorporated in their entirety herein by reference.

BACKGROUND

(1) Vascular lesions within and adjacent to vessels in the body can be associated with an increased risk for major adverse events, such as myocardial infarction, embolism, deep vein thrombosis, stroke, and the like. Severe vascular lesions can be difficult to treat and achieve patency for a physician in a clinical setting.

(2) Vascular lesions may be treated using interventions such as drug therapy, balloon angioplasty, atherectomy, stent placement, and vascular graft bypass, to name a few. Such interventions may not always be ideal or may require subsequent treatment to address the lesion.

(3) Creation of a plasma via optical breakdown of an aqueous solution requires a significant amount of energy in a short amount of time, upon which it is converted into a therapeutic bubble and/or a therapeutic pressure wave. With sufficiently high energy and short pulse durations, there is potential to damage a distal end of a light guide used to deliver light energy to generate the plasma. A means to enhance the conversion efficiency of the light energy to (plasma) pressure wave and bubble growth would reduce the required power handling requirements of the optical delivery

system. Therefore, less input energy would be required for an equivalent therapy while minimizing potential damage to the light guide.

(4) Creation of the plasma near the distal end of a small diameter light guide as in the case of aqueous optical breakdown as one method for an intravascular lithotripsy catheter has the potential for self-damage due to its proximity to the plasma creation and/or the pressure wave, high plasma temperatures, and waterjet from collapse of the bubble, as non-exclusive examples.

SUMMARY

(5) This summary is an overview of some of the teachings of the present application and is not intended to be an exclusive or exhaustive treatment of the present subject matter. Further details are found in the detailed description and appended claims. Other aspects will be apparent to persons skilled in the art upon reading and understanding the following detailed description and viewing the drawings that form a part thereof, each of which is not to be taken in a limiting sense. The scope herein is defined by the appended claims and their legal equivalents.

(6) The present invention is directed toward a catheter system for treating a treatment site within or adjacent to a blood vessel. In certain embodiments, the catheter system includes a power source, a light guide, and a plasma target. The light guide receives power from the power source. The light guide has a distal tip, and the light guide emits light energy in a direction away from the distal tip. The plasma target is spaced apart from the distal tip of the light guide by a target gap distance. The plasma target is configured to receive light energy from the light guide so that a plasma is generated at the plasma target upon receiving the light energy from the light guide.

(7) In some embodiments, the power source is a laser. In various embodiments, the light guide is an optical fiber.

(8) In certain embodiments, the catheter system can also include an inflatable balloon that encircles the distal tip of the light guide.

(9) In various embodiments, the catheter system can also include an inflatable balloon. In some such embodiments, the plasma target can be positioned within the inflatable balloon.

(10) In some embodiments, the target gap distance is greater than 1 μm , 10 μm , 100 μm , 1 mm, 2 mm, 3 mm, 5 mm, and/or 1 cm.

(11) In various embodiments, the plasma target can have a substantially circular cross-sectional configuration, a substantially square cross-sectional configuration, a substantially rectangular cross-sectional configuration, a substantially oval cross-sectional configuration, a substantially pentagonal cross-sectional configuration, a substantially hexagonal cross-sectional configuration, a substantially octagonal cross-sectional configuration, a polygonal cross-sectional configuration, a parallelogram cross-sectional configuration, a trapezoidal cross-sectional configuration or a substantially diamond-shaped cross-sectional configuration.

(12) In certain embodiments, the catheter system can also include a guidewire lumen. In some such embodiments, the light guide can be coupled to the guidewire lumen.

(13) In some embodiments, the plasma target has a target face that receives the light energy from the light guide. In various embodiments, the target face has an angle that is substantially orthogonal relative to a direction the light energy is emitted to the plasma target. In various embodiments, the target face has an angle that is greater than approximately 45 degrees and less than approximately 135 degrees relative to a direction the light energy is emitted to the plasma target. In certain embodiments, the target face can have an angle that is greater than zero degrees and less than 180 degrees relative to a direction the light energy is emitted to the plasma target.

(14) In various embodiments, the light guide includes a distal region having a longitudinal axis. The direction the light energy is emitted can be substantially along the longitudinal axis of the distal region. Alternatively, the direction the light energy is emitted can be substantially perpendicular to the longitudinal axis of the distal region. Still alternatively, the direction the light energy is emitted can be angled relative to the longitudinal axis of the distal region. For example, in some embodiments, the direction the light energy is emitted has an angle relative to the

longitudinal axis that is greater than zero degrees and less than 180 degrees. In various embodiments, the direction the light energy is emitted can have an angle relative to the longitudinal axis that is greater than 45 degrees and less than 135 degrees.

(15) In certain embodiments, the catheter system can include a plurality of plasma targets that are spaced apart from the distal tip of the light guide. In some such embodiments, at least one of the plurality of plasma targets can be configured to receive light energy from the light guide.

(16) In various embodiments, the plasma target can be at least partially formed from one of stainless steel and its variants, tungsten, tantalum, platinum, molybdenum, niobium, and iridium.

(17) In some embodiments, the plasma target can be at least partially formed from one of magnesium oxide, beryllium oxide, tungsten carbide, titanium nitride, titanium carbonitride, and titanium carbide.

(18) In certain embodiments, the plasma target can be at least partially formed from one of diamond CVD and diamond.

(19) In various embodiments, the plasma target can be at least partially formed from a transition metal, a metal alloy, and/or a ceramic material.

(20) In some embodiments, the plasma target can be fixedly coupled to the light guide.

Alternatively, the plasma target can be movably coupled to the light guide. Still alternatively, the plasma target can be uncoupled from the light guide.

(21) In some applications, the catheter system can include a guidewire lumen. In some such embodiments, the plasma target can substantially encircle the guidewire lumen.

(22) In certain embodiments, the target face can include one or more surface features, which can include one or more of an indentation, a projection, and a beveled edge.

(23) In some embodiments, the target face can have a conical configuration, a pyramidal configuration, a dome-shaped configuration, a concave configuration, a convex configuration, a multi-faceted configuration, a coiled configuration, a spring-like configuration, and/or a somewhat spiral configuration.

(24) In various embodiments, the plasma target can be movable relative to the light guide. In some embodiments, the plasma target can be spring-loaded.

(25) In certain embodiments, the catheter system can include a guidewire lumen, and the plasma target can be secured or otherwise coupled to the guidewire lumen.

(26) In some embodiments, the catheter system can include a second light guide that receives power from the power source. The second light guide can have a second distal tip. The second light guide can emit light energy in a direction away from the second distal tip toward the plasma target. The plasma target can be spaced apart from the second distal tip of the second light guide. The plasma target can be configured to receive light energy from the second light guide so that a second plasma is generated at the plasma target upon receiving the light energy from the second light guide.

(27) In certain embodiments, the catheter system can include a second light guide and a second plasma target. The second light guide can receive power from the power source. The second light guide can have a second distal tip. The second light guide can emit light energy in a direction away from the second distal tip toward the second plasma target. The second plasma target can be spaced apart from the plasma target and the second distal tip of the second light guide. The second plasma target can be configured to receive light energy from the second light guide so that a second plasma is generated at the second plasma target upon receiving the light energy from the second light guide.

(28) In various embodiments, the present invention can also be directed toward a method for creating plasma to optically break up vascular calcium in a blood vessel using laser pulse energy. In certain embodiments, the method includes the step of providing any one of the catheter systems shown and/or described herein.

(29) In some embodiments, the plasma target can be formed at least partially from a polymer, a polymeric material, and/or a plastic such as polyimide and nylon.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The novel features of this invention, as well as the invention itself, both as to its structure and its operation, will be best understood from the accompanying drawings, taken in conjunction with the accompanying description, in which similar reference characters refer to similar parts, and in which:
- (2) FIG. 1 is a schematic cross-sectional view of a catheter system having features of the present invention in accordance with various embodiments herein;
- (3) FIG. 2 is a simplified schematic side view of one embodiment of a portion of the catheter system, including one embodiment of a portion of a catheter;
- (4) FIG. 3A is a simplified schematic side view of one embodiment of a portion of the catheter system, including another embodiment of a portion of the catheter, shown in an inflated state;
- (5) FIG. 3B is a simplified schematic side view of the portion of the catheter illustrated in FIG. 3A, shown in a deflated state;
- (6) FIG. 4 is a simplified schematic side view of one embodiment of a portion of the catheter system, including another embodiment of a portion of the catheter;
- (7) FIG. 5 is a simplified schematic side view of one embodiment of a portion of the catheter system, including another embodiment of a portion of the catheter;
- (8) FIG. 6 is a simplified schematic side view of one embodiment of a portion of the catheter system, including another embodiment of a portion of the catheter;
- (9) FIG. 7A is a simplified schematic side view of one embodiment of a portion of the catheter system, including another embodiment of a portion of the catheter;
- (10) FIG. 7B is a simplified schematic side view of one embodiment of a portion of the catheter system, including another embodiment of a portion of the catheter;
- (11) FIG. 8 is a schematic cross-sectional view of the catheter system taken on line 8-8 in FIG. 1;
- (12) FIG. 9 is a schematic cross-sectional view of another embodiment of the catheter system;
- (13) FIG. 10 is a schematic cross-sectional view of yet another embodiment of the catheter system;
- (14) FIG. 11 is a schematic cross-sectional view of still another embodiment of the catheter system;
- (15) FIG. 12 is a schematic cross-sectional view of a portion of the catheter system including one embodiment of a distal portion of a light guide;
- (16) FIG. 13 is a schematic cross-sectional view of a portion of the catheter system including an embodiment of the distal portion of the light guide;
- (17) FIG. 14 is a schematic cross-sectional view of a portion of the catheter system including another embodiment of the distal portion of the light guide;
- (18) FIG. 15 is a schematic cross-sectional view of a portion of the catheter system including yet another embodiment of the distal portion of the light guide;
- (19) FIG. 16 is a simplified schematic side view of a portion of one embodiment of the catheter, including an embodiment of a portion of a plasma target;
- (20) FIGS. 16A-16J are cross-sectional views of various embodiments of the plasma target taken on line 16-16 in FIG. 16;
- (21) FIGS. 17A-17H are perspective views of various embodiments of a portion of the plasma target having a target face;
- (22) FIG. 18 is a cross-sectional view of a portion of the catheter system including one embodiment of a portion of the catheter;
- (23) FIG. 19 is a cross-sectional view of a portion of the catheter system including another

embodiment of a portion of the catheter;

(24) FIG. 20 is a cross-sectional view of a portion of the catheter system including another embodiment of a portion of the catheter; and

(25) FIG. 21 is a cross-sectional view of a portion of the catheter system including another embodiment of a portion of the catheter.

(26) While embodiments are susceptible to various modifications and alternative forms, specifics thereof have been shown by way of example and drawings, and will be described in detail. It should be understood, however, that the scope herein is not limited to the particular aspects described. On the contrary, the intention is to cover modifications, equivalents, and alternatives falling within the spirit and scope herein.

DESCRIPTION

(27) Treatment of vascular lesions can reduce major adverse events or death in affected subjects. A major adverse event is one that can occur anywhere within the body due to the presence of a vascular lesion. Major adverse events can include, but are not limited to major adverse cardiac events, major adverse events in the peripheral or central vasculature, major adverse events in the brain, major adverse events in the musculature, or major adverse events in any of the internal organs.

(28) In various embodiments, the systems and methods disclosed herein describe the use of a catheter systems including any number of light guides for generating pressure waves within an inflatable balloon (sometimes referred to herein simply as “balloon”) for disrupting intervascular lesions. The catheter systems herein can utilize light energy to generate a plasma near the light guide disposed in the inflatable balloon located at or near a treatment site. As used herein, the treatment site can include a vascular lesion such as a calcified vascular lesion or a fibrous vascular lesion (hereinafter sometimes referred to simply as a “lesion”), typically found in a blood vessel and/or a heart valve. The plasma formation can initiate a pressure wave and can initiate the rapid formation of one or more bubbles that can rapidly expand to a maximum size and then dissipate through a cavitation event that can also launch a pressure wave upon collapse. The rapid expansion of the plasma-induced bubbles can generate one or more pressure waves within a balloon fluid and thereby impart pressure waves upon the treatment site. The pressure waves can transfer mechanical energy through an incompressible balloon fluid to a treatment site to impart a fracture force on the lesion. Without wishing to be bound by any particular theory, it is believed that the rapid change in balloon fluid momentum upon a balloon wall of the inflatable balloon that is in contact with or positioned near the lesion is transferred to the lesion to induce fractures in the lesion.

(29) The catheter systems can include a catheter configured to advance to the lesion located within or adjacent to the blood vessel, where the catheters include a catheter shaft. The catheters also include one or more light guides disposed along the catheter shaft and within a balloon. Each light guide can be configured to be in optical communication with a light and/or power source.

(30) Those of ordinary skill in the art will realize that the following detailed description of the present invention is illustrative only and is not intended to be in any way limiting. Other embodiments of the present invention will readily suggest themselves to such skilled persons having the benefit of this disclosure. Additionally, other methods of delivering energy to the lesion can be utilized, including, but not limited to, electric current-induced plasma generation. Reference will now be made in detail to implementations of the present invention, as illustrated in the accompanying drawings.

(31) In the interest of clarity, not all of the routine features of the implementations described herein are shown and described. It will, of course, be appreciated that in the development of any such actual implementation, numerous implementation-specific decisions must be made in order to achieve the developer's specific goals, such as compliance with application-related and business-related constraints, and that these specific goals will vary from one implementation to another and from one developer to another. Moreover, it is appreciated that such a development effort might be

complex and time-consuming, but would nevertheless be a routine undertaking of engineering for those of ordinary skill in the art having the benefit of this disclosure.

(32) As an overview, in certain embodiments, the light guides can be configured to include one or more diverting features configured to direct light to exit from the light guide toward a side surface of the light guide and toward the balloon wall. The diverting features can direct light to exit in a direction away from the axis of the light guide, or in an off-axis direction. Additionally, or in the alternative, the light guides can each include one or more light windows disposed along the longitudinal or axial surfaces of each light guide and in optical communication with a diverting feature. The light windows can include a portion of the light guide that allows light to exit the light guide from within the light guide, such as a portion of the light guide lacking a cladding material on or about the light guide. The inflatable balloons described herein can be coupled to the catheter shaft and/or other structures, and can be inflated with a balloon fluid.

(33) The inflatable balloon can include a balloon wall and can be configured to expand from a deflated state suitable for advancing the catheter through a patient's vasculature to an inflated state suitable for anchoring the catheter in position relative to a treatment site. The power source can be configured to provide sub-millisecond pulses of a light from the power source to initiate plasma formation in a balloon fluid within the balloon to cause rapid bubble formation and to impart pressure waves upon the treatment site.

(34) Various embodiments of this invention shine laser light energy onto a plasma target to cause plasma generation via interaction with plasma target material rather than optical breakdown of the balloon fluid thereby moving the plasma creation away from a distal end of the optical fiber (light guide). This can be accomplished by positioning the plasma target away from the distal end of the optical fiber to absorb the light energy and convert it into a plasma at some distance away from the distal end of the light guide.

(35) As used herein, the terms “intravascular lesion” and “vascular lesion” are used interchangeably unless otherwise noted.

(36) It is appreciated that the catheter systems herein can include many different forms. Referring now to FIG. 1, a schematic cross-sectional view is shown of a catheter system in accordance with various embodiments herein. A catheter system **100** is suitable for imparting pressure to induce fractures in a vascular lesion within or adjacent a vessel wall of a blood vessel. In the embodiment illustrated in FIG. 1, the catheter system **100** can include one or more of a catheter **102**, one or more light guides **122**, a power source **124**, a manifold **136**, and a fluid pump **138**.

(37) The catheter **102** includes an inflatable balloon **104** (sometimes referred to herein as “balloon”). The catheter **102** is configured to move to a treatment site **106** within or adjacent to a blood vessel **108**. The treatment site **106** can include a vascular lesion such as a calcified vascular lesion, for example. Additionally, or in the alternative, the treatment site **106** can include a vascular lesion such as a fibrous vascular lesion.

(38) The catheter **102** can include the balloon **104**, a catheter shaft **110**, and a guidewire **112**. The balloon can be coupled to the catheter shaft **110**. The balloon can include a balloon proximal end **104P** and a balloon distal end **104D**. The catheter shaft **110** can extend between a shaft proximal end **114** and a shaft distal end **116**. The catheter shaft **110** can include a guidewire lumen **118** which is configured to move over the guidewire **112**. The catheter shaft **110** can also include an inflation lumen (not shown). In some embodiments, the catheter **102** can have a distal end opening **120** and can accommodate and be moved over and/or along the guidewire **112** so that the balloon **104** is positioned at or near the treatment site **106**.

(39) The catheter shaft **110** of the catheter **102** can encircle one or more light guides **122** (only one light guide **122** is illustrated in FIG. 1 for clarity) in optical communication with a power source **124**. The light guide **122** can be at least partially disposed along and/or within the catheter shaft **110** and at least partially within the balloon **104**. In various embodiments, the light guide **122** can be an optical fiber, and the power source **124** can be a laser. The power source **124** can be in optical

communication with the light guide **122**. In some embodiments, the catheter shaft **110** can encircle multiple light guides, such as a second light guide, a third light guide, etc.

(40) The balloon **104** can include a balloon wall **130**. The balloon **104** can expand from a collapsed configuration suitable for advancing at least a portion of the catheter shaft **102** through a patient's vasculature to an expanded configuration suitable for anchoring the catheter **102** into position relative to the treatment site **106**. The power source **124** of the catheter system **100** can be configured to provide sub-millisecond pulses of light from the power source **124**, along the light guide **112**, to a location within the balloon **104**. The pulses of light, resulting in light energy, thereby induce plasma formation in a balloon fluid **132** within the balloon **104**. The plasma formation causes rapid bubble formation, and imparts pressure waves upon the treatment site **106**. Exemplary plasma-induced bubbles are shown as bubbles **134** in FIG. 1. The balloon fluid **132** can be a liquid or a gas. As provided in greater detail herein, the plasma-induced bubbles **134** are intentionally formed at some distance away from the light guide **122** so that the likelihood of damage to the light guide is decreased.

(41) In various embodiments, the sub-millisecond pulses of light can be delivered to near the treatment site **106** at a frequency of from at least approximately 1 hertz (Hz) up to approximately 5000 Hz. In some embodiments, the sub-millisecond pulses of light can be delivered to near the treatment site **106** at a frequency from at least 30 Hz to 1000 Hz. In other embodiments, the sub-millisecond pulses of light can be delivered to near the treatment site **106** at a frequency from at least 10 Hz to 100 Hz. In yet other embodiments, the sub-millisecond pulses of light can be delivered to near the treatment site **106** at a frequency from at least 1 Hz to 30 Hz. In some embodiments, the sub-millisecond pulses of light can be delivered to near the treatment site **106** at a frequency that can be greater than or equal to 1 Hz, 2 Hz, 3 Hz, 4 Hz, 5 Hz, 6 Hz, 7 Hz, 8 Hz, or 9 Hz, 10 Hz, 20 Hz, 30 Hz, 40 Hz, 50 Hz, 60 Hz, 70 Hz, 80 Hz, 90 Hz, 100 Hz, 200 Hz, 300 Hz, 400 Hz, 500 Hz, 600 Hz, 700 Hz, 800 Hz, 900 Hz, 1000 Hz, 1250 Hz, 1500 Hz, 1750 Hz, 2000 Hz, 2250 Hz, 2500 Hz, 2750 Hz, 3000 Hz, 3250 Hz, 3500 Hz, 3750 Hz, 4000 Hz, 4250 Hz, 4500 Hz, 4750 Hz, or 5000 Hz or can be an amount falling within a range between any of the foregoing. Alternatively, the sub-millisecond pulses of light can be delivered to near the treatment site **106** at a frequency that can be greater than 5000 Hz.

(42) It is appreciated that the catheter system **100** herein can include any number of light guides **122** in optical communication with the power source **124** at the proximal portion **114**, and with the balloon fluid **132** within the balloon **104** at the distal portion **116**. For example, in some embodiments, the catheter system **100** herein can include from one light guide **122** to five light guides **122**. In other embodiments, the catheter system **100** herein can include from five light guides to fifteen light guides. In yet other embodiments, the catheter system **100** herein can include from ten light guides to thirty light guides. The catheter system **100** herein can include 1-30 light guides. It is appreciated that the catheter system **100** herein can include any number of light guides that can fall within a range, wherein any of the foregoing numbers can serve as the lower or upper bound of the range, provided that the lower bound of the range is a value less than the upper bound of the range. In some embodiments, the catheter system **100** herein can include greater than 30 light guides.

(43) The manifold **136** can be positioned at or near the shaft proximal end **114**. The manifold **136** can include one or more proximal end openings that can receive the one or more light guides, such as light guide **122**, the guidewire **112**, and/or an inflation conduit **140**. The catheter system **100** can also include the fluid pump **138** that is configured to inflate the balloon **104** with the balloon fluid **132** and/or deflate the balloon **104** as needed.

(44) As with all embodiments illustrated and described herein, various structures may be omitted from the figures for clarity and ease of understanding. Further, the figures may include certain structures that can be omitted without deviating from the intent and scope of the invention.

(45) FIG. 2 is a simplified schematic side view of one embodiment of a portion of the catheter

system **200**, including one embodiment of a portion of a catheter **202**. In the embodiment illustrated in FIG. 2, the catheter system can include one or more of an inflatable balloon **204**, a guidewire lumen **218**, and a light guide **222**. Although the light guide **222** in FIG. 2 (as well as other embodiments shown and/or described herein) is illustrated as being positioned adjacent to the guidewire lumen **218**, it is understood that in some embodiments, the light guide **222** can be positioned within the guidewire lumen **218** or the catheter shaft **110** (illustrated in FIG. 1), or the light guide **222** can be incorporated into a portion of the guidewire lumen **218** or the catheter shaft **110**. In still other embodiments, the light guide **222** can be positioned away from the guidewire lumen **218** and/or the catheter shaft **110**. In yet other embodiments, the guidewire lumen **218** can be omitted from the catheter system **200**. It is further recognized that the structures with reference characters included in FIG. 2 (as well as other figures shown and described herein) are not necessarily drawn to scale for ease of viewing and/or understanding.

(46) In the embodiment illustrated in FIG. 2, the catheter system **200** also includes a plasma target **242** that is spaced apart from the distal tip **244** of the light guide **222**. The plasma target **242** can be formed from various materials. In some embodiments, the plasma target **242** can be formed from metallics and/or metal alloys having relatively high melting temperatures, such as tungsten, tantalum, molybdenum, niobium, platinum and/or iridium. Alternatively, the plasma target **242** can be formed from at least one of magnesium oxide, beryllium oxide, tungsten carbide, titanium nitride, titanium carbonitride, and titanium carbide. Still alternatively, the plasma target **242** can be formed from at least one of diamond CVD and diamond. In other embodiments, the plasma target **242** can be formed from a transition metal, an alloy metal, or a ceramic material. Yet alternatively, in some embodiments, plasma target **242** can be formed at least partially from a polymer, a polymeric material, and/or a plastic such as polyimide and nylon. Still alternatively, the plasma target **242** can be formed from any other suitable material(s). As provided in greater detail herein, the geometry, configuration, size, and/or shape of the plasma target **242** can also be varied to suit the design requirements of the catheter system **200**.

(47) In the embodiment illustrated in FIG. 2, the light guide **222** emits light energy **243** (illustrated in dashed lines in FIG. 2) from a distal tip **244** of the light guide **222** toward the plasma target **242**. The plasma target **242** is spaced apart from the distal tip **244** of the light guide by a target gap distance **245**. The target gap distance **245** can vary. For example, in various embodiments, the target gap distance **245** can be at least 1 μm , at least 10 μm , at least 100 μm , at least 1 mm, at least 2 mm, at least 3 mm, at least 5 mm or at least 1 cm. The target gap distance **245** can vary depending upon the size, shape, and/or angle of the plasma target **242** relative to the light energy emitted by the light guide **222**, the type of material used to form the plasma target **242**, the quantity and/or duration of the light energy being emitted from the light guide **222**, the type of balloon fluid **232** used in the balloon **204**, etc.

(48) In certain embodiments, the plasma target **242** can be secured to another structure of the catheter system **200**. For example, the plasma target **242** can be fixedly or movably secured or coupled to the guidewire lumen **218**, as illustrated in FIG. 2. Alternatively, the plasma target **242** can be fixedly or movably secured or coupled to the light guide **222** or another suitable structure. Still, alternatively, the plasma target **242** can be suspended (unsecured) within the balloon fluid **232**.

(49) With this design, the light energy **243** generates a plasma bubble **234**, which creates an outwardly emanating pressure wave (not shown) throughout the balloon fluid **232** that impacts the balloon **204**. The impact to the balloon **204** causes the balloon to forcefully disrupt and/or fracture the vascular lesion, e.g., a calcified vascular lesion, at the treatment site **106** (illustrated in FIG. 1). In other words, the associated rapid formation of the plasma bubble **234** and resulting localized balloon fluid **232** velocity within the balloon **204** transfers mechanical energy through the incompressible balloon fluid **232** to impart a fracture force on the treatment site **106**. The rapid change in momentum of the balloon fluid **232** upon hitting the balloon wall **230** is known as

hydraulic shock, or water hammer. The change in momentum of the balloon fluid 232 is transferred as a fracture force to the vascular lesion, which is opposed to the balloon wall 230.

(50) By positioning the plasma target 242 away from the distal tip 244 of the light guide 222, damage to the light guide 222 from the plasma bubble 234 is less likely to occur than if the plasma bubble 234 was generated at or more proximate to the distal tip 244 of the light guide. Stated another way, the presence of the plasma target 242, and positioning the plasma target 242 away from the distal tip 244 of the light guide 222, causes the plasma bubble 234 to, in turn, be generated away from the distal tip 244 of the light guide 222, reducing the likelihood of damage to the light guide 222. Further, in this embodiment, the positioning of the plasma target 242 can also be different from those previously described.

(51) FIG. 3A is a simplified schematic side view of another embodiment of a portion of the catheter system 300, including another embodiment of a portion of the catheter 302, shown in an inflated state. In this embodiment, the catheter 302 includes a balloon 304, a guidewire lumen 318, one or more light guides 322 (two light guides 322 are illustrated in FIG. 3A), and one or more plasma targets 342 (two plasma targets 342 are illustrated in FIG. 3A). In the embodiment illustrated in FIG. 3A, the light guides 322 can be substantially similar to the light guides previously shown and described herein and/or shown in greater detail below.

(52) However, in this embodiment, the plasma targets 342 can be movable depending upon the inflation status of the balloon 304. For example, the plasma targets 342 can include springs, e.g., can be spring-loaded, that extend outwardly toward the balloon 304 when the balloon 304 is in the inflated state. Stated another way, the plasma targets 342 can move and/or extend toward the balloon 304 (or in another suitable direction) so that the light energy 343 from the light guide(s) 322 is better directed toward the plasma target(s) 342. As previously described herein, because the plasma target 342 is positioned away from the distal tip 344 of the light guide 322, the plasma bubble 334 that is generated is less likely to cause damage to the light guide than if the plasma bubble 334 were generated at or more near to the light guide 322.

(53) Further, in this embodiment, the positioning of the plasma targets 342 can be staggered (with two or more plasma targets 342) so that a greater area of the balloon 304 can be impacted by the resultant pressure wave(s) from the plasma bubbles 334.

(54) FIG. 3B is a simplified schematic side view of the portion of the catheter illustrated in FIG. 3A, shown in a deflated state. In this embodiment, upon deflation of the balloon 304, the plasma targets 342 can retract or otherwise move back toward the guidewire lumen 318, or in another suitable direction, so that the catheter 302 can have a somewhat smaller diameter during insertion and/or removal of the catheter 302 from the blood vessel 108 (illustrated in FIG. 1), thereby increasing the ease of insertion and/or removal by the operator of the catheter system 300.

(55) FIG. 4 is a simplified schematic side view of another embodiment of a portion of the catheter system 400, including another embodiment of a portion of the catheter 402. In the embodiment illustrated in FIG. 4, the catheter 402 includes a balloon 404, a guidewire lumen 418, one or more light guides 422, and one or more plasma targets 442.

(56) The operation and function of the light guide 422 and the plasma target 442 can be substantially similar to those previously described. However, in this embodiment, the light guide 422 can be configured to redirect the light energy 443 in a different direction, i.e., non-parallel with a longitudinal axis 470 of the light guide 422. For example, the light energy can be redirected at an angle α relative to the longitudinal axis 470 of the light guide 422. In the embodiment illustrated in FIG. 4, the light energy 443 is redirected in a direction that is somewhat perpendicular to the longitudinal axis 470 of the light guide 422. However, it is understood that this type of angle is provided for ease of understanding only, and that any angle α between 0 and 180 degrees relative to the longitudinal axis 470 of the light guide 422 can be used. The structures and methods for redirecting the light energy 443 in this manner are provided in greater detail herein.

(57) Further, in this embodiment, the positioning of the plasma target 442 can also be different

from those previously described. For example, in one embodiment, the plasma target **442** is positioned between the light guide **422** and the balloon **404**. In various embodiments, the plasma target **442** can be secured or coupled to another structure within the catheter **402**, such as the guidewire lumen **418**, the light guide **422**, the balloon **404**, or any other suitable structure. With this design, the plasma bubble **434** can be generated more proximate to the balloon **404**, which can be beneficial for exerting a greater force to disrupt and/or fracture the calcified lesion and/or to maintain a spacing between the formation of the plasma bubble **434** and the light guide **422** for reasons provided herein.

(58) FIG. 5 is a simplified cross-sectional view of another embodiment of a portion of the catheter system **500**, including another embodiment of a portion of the catheter **502**. In the embodiment illustrated in FIG. 5, the balloon has been omitted for clarity. In this embodiment, the catheter **502** includes a guidewire lumen **518**, one or more light guides **522**, one or more plasma targets **542**, and a target coupler **571**.

(59) The operation and function of the light guide **522** and the plasma target **542** can be substantially similar to those previously described. However, in this embodiment, the plasma target **542** is coupled to the guidewire lumen **518** (or another suitable structure) with the target coupler **571**. In one embodiment, the target coupler **571** can be a ring-like structure that secures the plasma target **542** to the guidewire lumen **518** (or another structure). The light energy **543** is emitted from the light guide **522**, and results in a plasma bubble **534** being generated at the plasma target **542**. Alternatively, the plasma target **542** can be adhered directly to the guidewire lumen **518** (or another structure) with adhesive or any other means for securing the plasma target **542**. Still alternatively, the target coupler **571** can be movable so that the plasma target **542** can be moved either manually or automatically along the guidewire lumen **518** to change the target gap distance **545** between the plasma target **542** and the distal tip **544** of the light guide **522**.

(60) Additionally, or in the alternative, the shape of the plasma target **542** can vary. For example, in the embodiment illustrated in FIG. 5, the plasma target **542** can have a somewhat dome-shape or convex configuration. Still, alternatively, the plasma target **542** can have another suitable configuration. With these designs, the plasma bubble **534** can be generated at the plasma target **542**, and the plasma target **542** can redirect the resultant pressure wave in any desired direction to achieve the desired results.

(61) FIG. 6 is a simplified schematic side view of another embodiment of a portion of the catheter system **600**, including another embodiment of a portion of the catheter **602**. In the embodiment illustrated in FIG. 6, the catheter **602** includes a balloon **604**, a guidewire lumen **618**, one or more light guides **622**, and one or more plasma targets **642**.

(62) The operation and function of the light guide **622** and the plasma target **642** can be substantially similar to those previously described. However, in this embodiment, the plasma target **642** is secured to a balloon inner surface **672** of the balloon **604**. With this design, the plasma bubble is generated away from the light guide **622**, thereby decreasing the likelihood of damage to the light guide **622**. Moreover, because the plasma target **642** is positioned on the balloon inner surface **672**, the balloon will receive a near-direct force from the plasma bubble **634** to increase the disruptive force upon the calcified lesion.

(63) In this embodiment, the light guide **622** can be configured to redirect the light energy **643** in a different direction, i.e., non-parallel with a longitudinal axis **670** of the light guide **622**. For example, the light energy can be redirected at an angle α relative to the longitudinal axis **670** of the light guide **622**. In the embodiment illustrated in FIG. 6, the light energy **643** is redirected in a direction that is somewhat perpendicular to the longitudinal axis **670** of the light guide **622**.

However, it is understood that this type of angle is provided for ease of understanding only, and that any angle α between 0 and 180 degrees relative to the longitudinal axis **670** of the light guide **622** can be used. The structures and methods for redirecting the light energy **643** in this manner are provided in greater detail herein.

(64) Further, in this embodiment, the positioning of the plasma target **642** can also be different from those previously described. For example, in one embodiment, the plasma target **642** is positioned between the light guide **622** and the balloon **604**. In various embodiments, the plasma target **642** can be secured or coupled to another structure within the catheter **602**, such as the guidewire lumen **618**, the light guide **622**, the balloon **604**, or any other suitable structure. With this design, the plasma bubble **634** can be generated more proximate to the balloon **604**, which can be beneficial for exerting a greater force to disrupt and/or fracture the calcified lesion and/or to maintain a spacing between the formation of the plasma bubble **634** and the light guide **622** for reasons provided herein.

(65) FIG. 7A is a simplified schematic side view of another embodiment of a portion of the catheter system **700A**, including another embodiment of a portion of the catheter **702A**. In the embodiment illustrated in FIG. 7A, the catheter **702A** includes a balloon **704**, a guidewire lumen **718**, one or more light guides **722A**, and a plurality of plasma targets **742**.

(66) The operation and function of the light guide **722A** and the plasma targets **742** can be substantially similar to those previously described. However, in this embodiment, the light guide **722A** can be configured to direct the light energy **743A** parallel with a longitudinal axis **770** of the light guide **722B** to generate plasma bubbles **734A** at a plurality of plasma targets **742**. In this embodiment, the plasma targets **742** are distributed throughout the balloon fluid **732** (illustrated as “X”s in FIG. 7A). In this embodiment, the plasma targets **742** can be relatively small so that they can be suspended in the balloon fluid **732** more readily. In various embodiments, the plasma targets **742** can be free-floating within the balloon fluid **732**, either as a homogeneous solution or a heterogeneous solution. With this design, one or more plasma bubbles **734A** (only one plasma bubble **734A** is illustrated in FIG. 7A) can be generated, which can be beneficial for exerting a greater force to disrupt and/or fracture the calcified lesion and/or to maintain a spacing between the formation of the plasma bubble **734A** and the light guide **722A** for reasons provided herein.

(67) FIG. 7B is a simplified schematic side view of another embodiment of a portion of the catheter system **700B**, including another embodiment of a portion of the catheter **702B**. In the embodiment illustrated in FIG. 7B, the catheter **702B** includes a balloon **704**, a guidewire lumen **718**, one or more light guides **722B**, and a plurality of plasma targets **742**.

(68) The operation and function of the light guide **722B** and the plasma targets **742** can be substantially similar to those previously described. However, in this embodiment, the light guide **722B** can be configured to redirect the light energy **743B** in a different direction, i.e., non-parallel with a longitudinal axis **770** of the light guide **722B**. For example, the light energy **743B** can be redirected at an angle α relative to the longitudinal axis **770** of the light guide **722B**. In the embodiment illustrated in FIG. 7B, the light energy **743B** is redirected in a direction that is somewhat perpendicular or orthogonal to the longitudinal axis **770** of the light guide **722B**. However, it is understood that this type of angle is provided for ease of understanding only, and that any angle α between 0 and 180 degrees relative to the longitudinal axis **770** of the light guide **722B** can be used. The structures and methods for redirecting the light energy **743B** in this manner are provided in greater detail herein.

(69) Further, in this embodiment, the positioning of the plasma targets **742** can also be different from those previously described. For example, in one embodiment, the plasma targets **742** are distributed throughout the balloon fluid **732** (illustrated as “X”s in FIG. 7B). In this embodiment, the plasma targets **742** can be relatively small so that they can be suspended in the balloon fluid **732** more readily. In various embodiments, the plasma targets **742** can be free-floating within the balloon fluid **732**, either as a homogeneous solution or a heterogeneous solution. With this design, one or more plasma bubbles **734B** (only one plasma bubble **734B** is illustrated in FIG. 7B) can be generated more proximate to the balloon **704**, which can be beneficial for exerting a greater force to disrupt and/or fracture the calcified lesion and/or to maintain a spacing between the formation of the plasma bubble **734B** and the light guide **722B** for reasons provided herein.

(70) Examples of the catheters in accordance with the various embodiments herein include those having multiple light guides disposed about the catheter shaft at different positions around the circumference, as shown in FIGS. **8-11**. It is understood that multiple light guides can be used with any of the embodiments shown and/or described herein without deviating from the intent and/or scope of the invention.

(71) Referring now to FIG. **8**, a schematic cross-sectional view of a catheter **102** in FIG. **1** along line **8-8** in FIG. **1** is shown in accordance with various embodiments herein. The catheter **802** illustrated in FIG. **8** can include one or more of a catheter shaft **810**, a guidewire **812**, a guidewire lumen **818**, a first light guide **822A** and a second light guide **822B** separated by about 180 degrees around the circumference from the first light guide **822A**. The first light guide **822A** includes a side surface that can include any surface portion about a circumference of the first light guide **822A**. The second light guide **822B** includes a side surface that can include any surface portion about the circumference of the second light guide **822B**. In some embodiments, the side surface spans a portion of the circumference of the light guides herein, such that it is less than cylindrical. In other embodiments, the side surface can span the entire circumference of the light guides herein such that it is cylindrical. It is recognized that any light guide described herein can include a side surface about the circumference of the light guide.

(72) Referring now to FIGS. **9-11**, schematic cross-sectional views of additional configurations for catheters having multiple light guides are shown in accordance with various embodiments herein. The embodiment of the catheter **902** illustrated in FIG. **3** can include one or more of a catheter shaft **910**, a guidewire **912**, a guidewire lumen **918**, a first light guide **922A**, a second light guide **922B**, and a third light guide **922C** separated by about 120 degrees around the circumference.

(73) The embodiment of the catheter **1002** illustrated in FIG. **10** includes one or more of a catheter shaft **1010**, a guidewire **1012**, a guidewire lumen **1018**, a first light guide **1022A**, a second light guide **1022B**, a third light guide **1022C**, and a fourth light guide **1022D** separated by about 90 degrees around the circumference.

(74) The embodiment of the catheter **1102** illustrated in FIG. **11** includes one or more of a catheter shaft **1110**, a guidewire **1112**, a guidewire lumen **1118**, a first light guide **1122A**, a second light guide **1122B**, a third light guide **1122C**, a fourth light guide **1122D**, a fifth light guide **1122E**, and a sixth light guide **1122F** separated by about 60 degrees around the circumference. It is understood that greater than six light guides can be used in the embodiments herein.

(75) It is further appreciated that the light guides described herein can be disposed uniformly or nonuniformly about the catheter shaft to achieve the desired effect in the desired locations.

(76) Diverting features and focusing features (also sometimes referred to herein simply as “diverting features”) will be discussed in more detail below and in reference to FIGS. **12-15**. The light guides herein can include one or more diverting features, where each diverting feature can be in optical communication with the light guide within which it is disposed. In some embodiments, the diverting features can be in optical communication with a distal end of the light guide. Referring now to FIGS. **12-15**, schematic cross-sectional views of the distal ends of various shaped light guides are shown in accordance with various embodiments herein.

(77) In FIG. **12**, a schematic cross-sectional view of a light guide **1222** is shown. Light guide **1222** is configured such that light **1254** travels from the power source **124** (illustrated in FIG. **1**) in a direction from the shaft proximal end **114** (illustrated in FIG. **1**) to the distal tip **1244**, as indicated by arrows **1254**.

(78) In some embodiments, the end of the light guide can have an angled shape. By way of example, in FIG. **13** a schematic cross-sectional view of a light guide **1322** is shown.

(79) In some embodiments, the end of the light guide can have a tapered shape. By way of example, in FIG. **14** a schematic cross-sectional view of a light guide **1422** is shown.

(80) Referring now to FIG. **15**, a schematic cross-sectional view of a light guide **1522** is shown. Light guide **1522** includes an angled end **1558** disposed on a side surface **1562** of a distal end **1564**

of the light guide **1522**. The light guide **1522** includes a diverting feature **1566** at the distal tip **1544** to direct the light energy **1554** within the light guide **1522** toward the side surface **1562** of the light guide **1522**. Light guide **1522** is configured such that light energy **1554** travels from the distal tip **1544** in a direction that is approximately 90 degrees (or another suitable angle) from the longitudinal axis **470** (illustrated in FIG. 4, for example) as indicated by arrows **1568**. Upon contact with the diverting feature **1566**, the light energy **1554** is diverted, or reflected, within the light guide **1522** to a side surface **1562** of the light guide **1522**. The light energy **1554** extends away from the side surface **1562** of the light guide **1522**.

(81) The diverting feature **1566** of light guide **1522** can be made from a reflecting element or a refracting element. The diverting feature **1566** can be made from a glass, a polymer, a mirror, or a reflective metal coating. It is appreciated that the angle of internal reflection by the diverting feature **1566** can be adjusted by changing the angle of the distal tip **1544** of light guide **1522**.

(82) In some embodiments, a diverting feature can be included with the light guide to direct light toward a side surface of the distal end of the light guide. A diverting feature can include any feature of the system herein that diverts light from the light guide away from its axial path toward a side surface of the light guide. Examples include a reflector, a refracting structure, and a fiber diffuser.

(83) In some embodiments herein, the light guides can include multiple diverting features. By way of example, each light guide herein can include a first diverting feature, a second diverting feature, a third diverting feature or a fourth diverting feature. In other embodiments, each light guide can include more than four diverting features. The diverting features can be configured to direct light to exit a light guide at a side surface thereof toward the balloon wall. In some examples, the diverting feature directs light toward the balloon surface closest to the diverting feature, so that the light does not cross the longitudinal axis of the catheter on its path to the balloon surface. It is appreciated that the diverting features can be in optical communication with corresponding light window.

(84) The diverting features herein can be configured to direct light in the light guide toward a side surface of the distal portion, where the side surface is in optical communication with a light window. It is appreciated that the light guides herein can each include multiple diverting features and multiple light windows. Examples of the diverting features suitable for use herein include a reflecting element, a refracting element and/or a fiber diffuser.

(85) FIG. 16 is a simplified schematic side view of one embodiment of a portion of a catheter, including a light guide **1622** and an embodiment of a portion of a plasma target **1642**. The plasma target **1642** includes a target face **1672**. The target face **1672** can have any suitable geometry, shape or configuration. In this embodiment, the light guide **1622** and the plasma target **1642** operate substantially similar to those previously shown and/or described. However, in this embodiment, the target face **1672** is angled relative to a longitudinal axis **1670** of the light guide **1622**. Stated another way, the target face **1672** has an angle α that can be any angle between 0 and 180 degrees relative to the longitudinal axis **1670** of the light guide **1622**.

(86) FIGS. 16A-16J are cross-sectional views showing representative, non-exclusive, non-limiting embodiments of the cross-sectional shape of the plasma target **1642**. It is understood that there are literally an infinite number of possible cross-sectional configurations for the plasma target **1642** and that it would be an impossibility to show and describe all such configurations. However, it is the intent that the scope of this invention would encompass all such potential configurations, even those that are not shown and/or described herein.

(87) FIG. 16A is a cross-sectional view of one embodiment of the plasma target **1642** taken on line 16-16 in FIG. 16. In this embodiment the plasma target **1642** has a substantially circular cross-sectional shape.

(88) FIG. 16B is a cross-sectional view of another embodiment of the plasma target **1642** taken on line 16-16 in FIG. 16. In this embodiment the plasma target **1642** has a substantially vertical oval or elliptical cross-sectional shape.

(89) FIG. 16C is a cross-sectional view of another embodiment of the plasma target **1642** taken on

line **16-16** in FIG. **16**. In this embodiment the plasma target **1642** has a substantially square cross-sectional shape.

(90) FIG. **16D** is a cross-sectional view of another embodiment of the plasma target **1642** taken on line **16-16** in FIG. **16**. In this embodiment the plasma target **1642** has a substantially diamond, trapezoidal or parallelogram cross-sectional shape.

(91) FIG. **16E** is a cross-sectional view of another embodiment of the plasma target **1642** taken on line **16-16** in FIG. **16**. In this embodiment the plasma target **1642** has a substantially hexagonal cross-sectional shape.

(92) FIG. **16F** is a cross-sectional view of another embodiment of the plasma target **1642** taken on line **16-16** in FIG. **16**. In this embodiment the plasma target **1642** has a substantially horizontal oval or elliptical cross-sectional shape.

(93) FIG. **16G** is a cross-sectional view of another embodiment of the plasma target **1642** taken on line **16-16** in FIG. **16**. In this embodiment the plasma target **1642** has a substantially pentagonal cross-sectional shape.

(94) FIG. **16H** is a cross-sectional view of another embodiment of the plasma target **1642** taken on line **16-16** in FIG. **16**. In this embodiment the plasma target **1642** has a substantially octagonal cross-sectional shape.

(95) FIG. **16I** is a cross-sectional view of another embodiment of the plasma target **1642** taken on line **16-16** in FIG. **16**. In this embodiment the plasma target **1642** has a substantially vertical rectangular cross-sectional shape.

(96) FIG. **16J** is a cross-sectional view of another embodiment of the plasma target **1642** taken on line **16-16** in FIG. **16**. In this embodiment the plasma target **1642** has a substantially horizontal rectangular cross-sectional shape.

(97) FIGS. **17A-17H** are perspective views showing representative, non-exclusive, non-limiting embodiments of the geometry, shape and/or configuration of the target face **1772A-H** of the plasma target **1742A-H**. It is understood that there are literally an infinite number of possible cross-sectional configurations for the target face **1772A-H** of the plasma target **1742A-H** and that it would be an impossibility to show and describe all such configurations. However, it is the intent that the scope of this invention would encompass all such potential configurations, even those that are not shown and/or described herein.

(98) FIG. **17A** is a perspective view of a portion of an embodiment of the plasma target **1742A** having one embodiment of a target face **1772A**. In this embodiment, the target face **1772A** has a somewhat conical configuration.

(99) FIG. **17B** is a perspective view of a portion of an embodiment of the plasma target **1742B** having one embodiment of a target face **1772B**. In this embodiment, the target face **1772B** has a somewhat pyramidal configuration.

(100) FIG. **17C** is a perspective view of a portion of an embodiment of the plasma target **1742C** having one embodiment of a target face **1772C**. In this embodiment, the target face **1772C** has a somewhat convex or dome-shaped configuration.

(101) FIG. **17D** is a perspective view of a portion of an embodiment of the plasma target **1742D** having one embodiment of a target face **1772D**. In this embodiment, the target face **1772D** has a somewhat concave configuration.

(102) FIG. **17E** is a perspective view of a portion of an embodiment of the plasma target **1742E** having one embodiment of a target face **1772E**. In this embodiment, the target face **1772E** includes a spiral projection **1774** that extends outwardly from a side portion **1776** of the plasma target **1742E**.

(103) FIG. **17F** is a perspective view of a portion of an embodiment of the plasma target **1742F** having one embodiment of a target face **1772F**. In this embodiment, the target face **1772F** has a somewhat spring-like or coiled configuration.

(104) FIG. **17G** is a perspective view of a portion of an embodiment of the plasma target **1742G**

having one embodiment of a target face **1772G**. In this embodiment, the target face **1772G** has a beveled configuration.

(105) FIG. **17H** is a perspective view of a portion of an embodiment of the plasma target **1742H** having one embodiment of a target face **1772H**. In this embodiment, the target face **1772H** includes one or more surface features **1778**. The surface features **1778** can include dimples, depressions or indentations that extend into a target surface **1780** of the target face **1772H**. Additionally, or in the alternative, the surface features **1778** can include projections that extend outwardly from the target surface **1780** of the target face **1772H**. In one embodiment, the surface features **1778** can include the same or other materials that are added to the target surface **1780**. The specific sizes and/or shape(s) of the surface features **1778** can be varied.

(106) FIG. **18** is a cross-sectional view of a portion of the catheter system **1800** including one embodiment of a portion of the catheter **1802**. In this embodiment, the catheter **1802** can include a guidewire lumen **1818**, one or more light guides **1822** and an embodiment of a portion of a plasma target **1842**. The plasma target **1842** is spaced apart from the light guide **1822**, and includes a target face **1872**. The target face **1872** can have any suitable geometry, shape or configuration. In this embodiment, the light guide **1822** and the plasma target **1842** operate substantially similar to those previously shown and/or described. However, in this embodiment, the plasma target **1842** can be annular and can encircle the circumference of the guidewire lumen **1818**. Further, the target face **1872** of the plasma target **1842** can have a somewhat concave, conical configuration. In alternative, non-exclusive embodiments, the plasma target **1842** can have a beveled, toroidal or frustoconical configuration, or any other suitable configuration. In an alternative embodiment, the plasma target **1842** only partially encircles the guidewire lumen **1818**. Still alternatively, the plasma target **1842** can encircle or partially encircle another structure of the catheter **1802**.

(107) FIG. **19** is a cross-sectional view of a portion of the catheter system **1900** including one embodiment of a portion of the catheter **1902**. In this embodiment, the catheter **1902** can include a guidewire lumen **1918**, one or more light guides **1922** and an embodiment of a portion of a plasma target **1942**. The plasma target **1942** is spaced apart from the light guide **1922**, and includes a target face **1972**. The target face **1972** can have any suitable geometry, shape or configuration. In this embodiment, the light guide **1922** and the plasma target **1942** operate substantially similar to those previously shown and/or described. However, in this embodiment, the plasma target **1942** can be annular and can encircle the circumference of the guidewire lumen **1918**. Further, the target face **1972** of the plasma target **1942** can have a somewhat conical or pyramidal configuration. In an alternative embodiment, the plasma target **1942** only partially encircles the guidewire lumen **1918**. Still alternatively, the plasma target **1942** can encircle or partially encircle another structure of the catheter **1902**.

(108) FIG. **20** is a cross-sectional view of a portion of the catheter system **2000** including one embodiment of a portion of the catheter **2002**. In this embodiment, the catheter **2002** can include a guidewire lumen **2018**, two or more light guides **2022** (only two light guides **2022** are illustrated in FIG. **20**) and an embodiment of a portion of a plasma target **2042**. The plasma target **2042** is spaced apart from the light guides **2022**, and includes a target face **2072**. The target face **2072** can have any suitable geometry, shape or configuration. In this embodiment, the light guides **2022** and the plasma target **2042** operate substantially similar to those previously shown and/or described. In an alternative embodiment, two or more plasma targets **2042** can be secured to the guidewire lumen **2018**, wherein each only partially encircles the guidewire lumen **2018**. Still alternatively, the plasma target **2042** can encircle or partially encircle another structure of the catheter **2002**. It is appreciated that greater than two light guides **2022** can be used with the catheter system **2000** herein. For example, three light guides **2022** can be evenly spaced apart by 120 degrees from one another; four light guides **2022** can be evenly spaced apart by 90 degrees from one another, etc. Still alternatively, any number of light guides **2022** may be positioned so they are not evenly spaced circumferentially around the guidewire lumen **2018**.

(109) FIG. 21 is a cross-sectional view of a portion of the catheter system **2100** including one embodiment of a portion of the catheter **2102**. In this embodiment, the catheter **2102** can include a guidewire lumen **2118**, two or more light guides including a first light guide **2122F**, and a second light guide **2122S** (only two light guides are illustrated in FIG. 21) and two or more plasma targets including a first plasma target **2142F** having a first target face **2172F**, and a second plasma target **2142S** (only two plasma targets are illustrated in FIG. 21) having a second target face **2172S**. In the embodiment illustrated in FIG. 21, the first light guide **2122F** emits light energy to generate a first plasma bubble **2134F** at or near the first target face **2172F** of the first plasma target **2142F**. The second light guide **2122S** emits light energy to generate a second plasma bubble **2134S** at or near the second target face **2172S** of the second plasma target **2142S**, which is spaced apart from the first plasma target **2142F**. In the embodiment illustrated in FIG. 21, the second light guide **2122S** extends through the first plasma target **2142F**. Alternatively, the second light guide **2122S** can traverse around the first plasma target **2142F**. Still alternatively, any or all of the plasma targets **2142F**, **2142S** can either encircle the guidewire lumen **2118**, or partially encircle the guidewire lumen **2118**.

(110) Balloons

(111) The balloons suitable for use in the catheter systems illustrated and/or described herein include those that can be passed through the vasculature of a patient when in a collapsed configuration. In some embodiments, the balloons illustrated and/or described herein are made from silicone. In other embodiments, the balloons herein are made from polydimethylsiloxane (PDMS), polyurethane, polymers such as PEBAX™ material available from Arkema, which has a location at King of Prussia, Pennsylvania, USA, nylon, and the like. In some embodiments, the balloons can include those having diameters ranging from 1 millimeter (mm) to 25 mm in diameter. In some embodiments, the balloons can include those having diameters ranging from at least 1.5 mm to 12 mm in diameter. In some embodiments, the balloons can include those having diameters ranging from at least 1 mm to 5 mm in diameter. In some embodiments, the diameter can be greater than or equal to 0.5 mm, 1.0 mm, 1.5 mm, 2.0 mm, 2.5 mm, 3.0 mm, 3.5 mm, 4.0 mm, 4.5 mm, 5.0 mm, 5.5 mm, 6.0 mm, 6.5 mm, 7.0 mm, 7.5 mm, 8.0 mm, 8.5 mm, 9.0 mm, 9.5 mm, 10.0 mm, 10.5 mm, 11.0 mm, 11.5 mm, 12.0 mm, 12.5 mm, 13.0 mm, 13.5 mm, 14.0 mm, 14.5 mm, 15.0 mm, 15.5 mm, 16.0 mm, 16.5 mm, 17.0 mm, 17.5 mm, 18.0 mm, 18.5 mm, 19.0 mm, 19.5 mm, or 20.0 mm, or can be an amount falling within a range between any of the foregoing.

(112) In some embodiments, the balloons illustrated and/or described herein can include those having a length ranging from at least 5 mm to 300 mm in length. In some embodiments, the balloons illustrated and/or described herein can include those having a length ranging from at least 8 mm to 200 mm in length. In some embodiments, the length of the balloon can be greater than or equal to 5 mm, 10 mm, 20 mm, 30 mm, 40 mm, 50 mm, 60 mm, 70 mm, 80 mm, 90 mm, 100 mm, 110 mm, 120 mm, 130 mm, 140 mm, 150 mm, 160 mm, 170 mm, 180 mm, 190 mm, 200 mm, 210 mm, 220 mm, 230 mm, 240 mm, 250 mm, 260 mm, 270 mm, 280 mm, 290 mm, or 300 mm, or can be an amount falling within a range between any of the foregoing.

(113) The balloons illustrated and/or described herein can be inflated to inflation pressures from 1 atmosphere (atm) to 70 atm. In some embodiments, the balloons illustrated and/or described herein can be inflated to inflation pressures of from at least 20 atm to 70 atm. In some embodiments, the balloons illustrated and/or described herein can be inflated to inflation pressures of from at least 6 atm to 20 atm. In some embodiments, the balloons illustrated and/or described herein can be inflated to inflation pressures of from at least 3 atm to 20 atm. In some embodiments, the balloons illustrated and/or described herein can be inflated to inflation pressures of from at least 2 atm to 10 atm. In some embodiments, the balloons illustrated and/or described herein can be inflated to inflation pressures that can be greater than or equal to 1 atm, 2 atm, 3 atm, 4 atm, 5 atm, 6 atm, 7 atm, 8 atm, 9 atm, 10 atm, 15 atm, 20 atm, 25 atm, 30 atm, 35 atm, 40 atm, 45 atm, 50 atm, 55 atm, 60 atm, 65 atm, or 70 atm, or can be an amount falling within a range between any of the

foregoing.

(114) The balloons illustrated and/or described herein can include those having various shapes, including, but not to be limited to, a conical shape, a square shape, a rectangular shape, a spherical shape, a conical/square shape, a conical/spherical shape, an extended spherical shape, an oval shape, a tapered shape, a bone shape, a stepped diameter shape, an offset shape, or a conical offset shape. In some embodiments, the balloons illustrated and/or described herein can include a drug eluting coating or a drug eluting stent structure. The drug elution coating or drug eluting stent can include one or more therapeutic agents including anti-inflammatory agents, anti-neoplastic agents, anti-angiogenic agents, and the like.

(115) Balloon Fluids

(116) Exemplary balloon fluids suitable for use herein can include, but are not to be limited to one or more of water, saline, contrast medium, fluorocarbons, perfluorocarbons, gases, such as carbon dioxide, and the like. In some embodiments, the balloon fluids illustrated and/or described herein can be used as base inflation fluids, discussed elsewhere herein. In some embodiments, the balloon inflation fluids include a mixture of saline to contrast medium in a volume ratio of 50:50. In some embodiments, the balloon fluids include a mixture of saline to contrast medium in a volume ratio of 25:75. In some embodiments, the balloon fluids include a mixture of saline to contrast medium in a volume ratio of 75:25. The balloon fluids suitable for use herein can be tailored on the basis of composition, viscosity, and the like in order to manipulate the rate of travel of the pressure waves therein. The balloon fluids suitable for use herein are biocompatible. A volume of balloon fluid can be tailored by the chosen power source and the type of balloon fluid used.

(117) In some embodiments, the contrast agents used in the contrast media herein can include, but are not to be limited to, iodine-based contrast agents, such as ionic or non-ionic iodine-based contrast agents. Some non-limiting examples of ionic iodine-based contrast agents include diatrizoate, metrizoate, iothalamate, and ioxaglate. Some non-limiting examples of non-ionic iodine-based contrast agents include iopamidol, iohexol, ioxilan, iopromide, iodixanol, and ioversol. In other embodiments, non-iodine based contrast agents can be used. Suitable non-iodine containing contrast agents can include gadolinium (III)-based contrast agents. Suitable fluorocarbon and perfluorocarbon agents can include, but are not to be limited to, agents such as the perfluorocarbon dodecafluoropentane (DDFP, C₅F₁₂).

(118) The balloon fluids illustrated and/or described herein can include those that include absorptive agents that can selectively absorb light in the ultraviolet (e.g., at least 10 nanometers (nm) to 400 nm), visible region (e.g., at least 400 nm to 780 nm), and near-infrared region of the electromagnetic spectrum (e.g., at least 780 nm to 2.5 μ m), or in the far-infrared region of the electromagnetic spectrum of at least 10 nm to 2.5 micrometers (μ m). Suitable absorptive agents can include those with absorption maxima along the spectrum from at least 10 nm to 2.5 μ m. In various embodiments, the absorptive agent can be those that have an absorption maximum matched with the emission maximum of the laser used in the catheter system. By way of non-limiting examples, various lasers described herein can include neodymium:yttrium-aluminum-garnet (Nd:YAG—emission maximum=1064 nm) lasers, holmium:YAG (Ho:YAG—emission maximum=2.1 μ m) lasers, or erbium:YAG (Er:YAG—emission maximum=2.94 μ m). In some embodiments, the absorptive agents used herein can be water soluble. In other embodiments, the absorptive agents used herein are not water soluble. In some embodiments, the absorptive agents used in the balloon fluids herein can be tailored to match the peak emission of the power source. Various power sources having emission wavelengths of at least 10 nanometers to 1 millimeter are discussed elsewhere herein.

(119) In some embodiments, introduction of the balloon fluid causes the expansion of the balloon from a collapsed configuration to a first expanded configuration and from a first expanded configuration to a second further expanded configuration. In addition, or alternatively, the expansion of the balloon can be accomplished using a shape-memory material or other means.

(120) Light Guides

(121) The light guides illustrated and/or described herein can include an optical fiber or flexible light pipe. The light guides illustrated and/or described herein can be thin and flexible and can allow light signals to be sent with very little loss of strength. The light guides illustrated and/or described herein can include a core surrounded by a cladding about its circumference. In some embodiments, the core can be a cylindrical core or a partially cylindrical core. The core and cladding of the light guides can be formed from one or more materials, including but not limited to one or more types of glass, silica, or one or more polymers. The light guides may also include a protective coating, such as a polymer. It is appreciated that the index of refraction of the core will be greater than the index of refraction of the cladding.

(122) Each light guide can guide light along its length to a distal portion having at least one optical window. The light guides can create a light path as portion of an optical network including a power source. The light path within the optical network allows light to travel from one part of the network to another. Both the optical fiber or the flexible light pipe can provide a light path within the optical networks herein.

(123) The light guides illustrated and/or described herein can assume many configurations about the catheter shaft of the catheters illustrated and/or described herein. In some embodiments, the light guides can run parallel to the longitudinal axis of the catheter shaft of the catheter. In some embodiments, the light guides can be disposed spirally or helically about the longitudinal axis of the catheter shaft of the catheter. In some embodiments, the light guides can be physically coupled to the catheter shaft. In other embodiments, the light guides can be disposed along the length of the outer diameter of the catheter shaft. In yet other embodiments the light guides herein can be disposed within one or more light guide lumens within the catheter shaft. Various configurations for the catheter shafts and light guide lumens will be discussed below.

(124) Diverting Features and Focusing Features

(125) The diverting features suitable for use herein include a reflecting element, a refracting element, and a fiber diffuser. In some embodiments, the diverting feature can be a reflecting element. In some embodiments, the diverting feature can be a refracting element. In some embodiments, the diverting feature can be a fiber diffuser.

(126) A fiber diffuser can direct light from within a light guide to exit at a side surface of the light guide. The fiber diffusers described herein can be created several ways. In some embodiments, the fiber diffusers can be created by micro-machining the surface of the distal portion of a light guide with a CO₂ laser. In some embodiments, a fused silica coating can be applied to the distal portion of the light guide. In other embodiments, the fiber diffuser can be formed from a glass, a polymer, or a metal coating on the distal portion of the light guide. In other embodiments, the fiber diffuser can be formed by a fiber Bragg grating on the distal portion of the light guide. In some embodiments, the fiber diffuser can include a machined portion of the light guide, a laser-machined portion of the light guide, fiber Bragg gratings, a fused splicing, a fused splicing forming at least one internal mirror, and a splicing of two or more diffuse regions.

(127) Suitable materials for a fiber diffuser can include, but are not be limited to, the materials of the light guide core or light guide cladding, ground glass, silver coated glass, gold coated glass, TiO₂, and other materials that will scatter and not significantly absorbed the light wavelength of interest. One method that can be used to create a uniform diffuser in a light guide, optical component, or materials is to utilize scattering centers on the order of at least 50 nanometers to 5 micrometers in size. The scattering centers can have a distribution about 200 nanometers in size.

(128) The diverting features and focusing features suitable for focusing light away from the tip of the light guides herein can include, but are not to be limited to, those having a convex surface, a gradient-index (GRIN) lens, and a mirror focus lens.

(129) Power Sources

(130) The power sources suitable for use herein can include various types of power sources

including lasers and lamps. Suitable lasers can include short pulse lasers on the sub-millisecond timescale. In some embodiments, the power source can include lasers on the nanosecond (ns) timescale. The lasers can also include short pulse lasers on the picosecond (ps), femtosecond (fs), and microsecond (us) timescales. It is appreciated that there are many combinations of laser wavelengths, pulse widths and energy levels that can be employed to achieve plasma in the balloon fluid of the catheters illustrated and/or described herein. In various embodiments, the pulse widths can include those falling within a range including from at least 10 ns to 200 ns. In some embodiments, the pulse widths can include those falling within a range including from at least 20 ns to 100 ns. In other embodiments, the pulse widths can include those falling within a range including from at least 1 ns to 5000 ns.

(131) Exemplary nanosecond lasers can include those within the UV to IR spectrum, spanning wavelengths of about 10 nanometers to 1 millimeter. In some embodiments, the power sources suitable for use in the catheter systems herein can include those capable of producing light at wavelengths of from at least 750 nm to 2000 nm. In some embodiments, the power sources can include those capable of producing light at wavelengths of from at least 700 nm to 3000 nm. In some embodiments, the power sources can include those capable of producing light at wavelengths of from at least 100 nm to 10 micrometers (μm). Nanosecond lasers can include those having repetition rates of up to 200 kHz. In some embodiments, the laser can include a Q-switched thulium:yttrium-aluminum-garnet (Tm:YAG) laser. In some embodiments, the laser can include a neodymium:yttrium-aluminum-garnet (Nd:YAG), holmium:yttrium-aluminum-garnet (Ho:YAG), erbium:yttrium-aluminum-garnet (Er:YAG), excimer laser, helium-neon laser, carbon dioxide laser, as well as doped, pulsed, fiber lasers.

(132) Pressure Waves

(133) The catheters illustrated and/or described herein can generate pressure waves having maximum pressures in the range of at least 1 megapascal (MPa) to 100 MPa. The maximum pressure generated by a particular catheter will depend on the power source, the absorbing material, the bubble expansion, the propagation medium, the balloon material, distance of measurement from plasma epicenter, and other factors. In some embodiments, the catheters illustrated and/or described herein can generate pressure waves having maximum pressures in the range of at least 2 MPa to 50 MPa. In other embodiments, the catheters illustrated and/or described herein can generate pressure waves having maximum pressures in the range of at least 2 MPa to 30 MPa. In yet other embodiments, the catheters illustrated and/or described herein can generate pressure waves having maximum pressures in the range of at least 15 MPa to 25 MPa. In some embodiments, the catheters illustrated and/or described herein can generate pressure waves having peak pressures of greater than or equal to 1 MPa, 2 MPa, 3 MPa, 4 MPa, 5 MPa, 6 MPa, 7 MPa, 8 MPa, 9 MPa, 10 MPa, 11 MPa, 12 MPa, 13 MPa, 14 MPa, 15 MPa, 16 MPa, 17 MPa, 18 MPa, 19 MPa, 20 MPa, 21 MPa, 22 MPa, 23 MPa, 24 MPa, or 25 MPa, 26 MPa, 27 MPa, 28 MPa, 29 MPa, 30 MPa, 31 MPa, 32 MPa, 33 MPa, 34 MPa, 35 MPa, 36 MPa, 37 MPa, 38 MPa, 39 MPa, 40 MPa, 41 MPa, 42 MPa, 43 MPa, 44 MPa, 45 MPa, 46 MPa, 47 MPa, 48 MPa, 49 MPa, or 50 MPa. It is appreciated that the catheters illustrated and/or described herein can generate pressure waves having operating pressures or maximum pressures that can fall within a range, wherein any of the foregoing numbers can serve as the lower or upper bound of the range, provided that the lower bound of the range is a value less than the upper bound of the range.

(134) Therapeutic treatment can act via a fatigue mechanism or a brute force mechanism. For a fatigue mechanism, operating pressures would be about at least 0.5 MPa to 2 MPa, or about 1 MPa. For a brute force mechanism, operating pressures would be about at least 20 MPa to 30 MPa, or about 25 MPa. Pressures between the extreme ends of these two ranges may act upon a treatment site using a combination of a fatigue mechanism and a brute force mechanism.

(135) The pressure waves described herein can be imparted upon the treatment site from a distance within a range from at least 0.1 millimeters (mm) to 25 mm extending radially from a longitudinal

axis of a catheter placed at a treatment site. In some embodiments, the pressure waves can be imparted upon the treatment site from a distance within a range from at least 10 mm to 20 mm extending radially from a longitudinal axis of a catheter placed at a treatment site. In other embodiments, the pressure waves can be imparted upon the treatment site from a distance within a range from at least 1 mm to 10 mm extending radially from a longitudinal axis of a catheter placed at a treatment site. In yet other embodiments, the pressure waves can be imparted upon the treatment site from a distance within a range from at least 1.5 mm to 4 mm extending radially from a longitudinal axis of a catheter placed at a treatment site. In some embodiments, the pressure waves can be imparted upon the treatment site from a range of at least 2 MPa to 30 MPa at a distance from 0.1 mm to 10 mm. In some embodiments, the pressure waves can be imparted upon the treatment site from a range of at least 2 MPa to 25 MPa at a distance from 0.1 mm to 10 mm. In some embodiments, the pressure waves can be imparted upon the treatment site from a distance that can be greater than or equal to 0.1 mm, 0.2 mm, 0.3 mm, 0.4 mm, 0.5 mm, 0.6 mm, 0.7 mm, 0.8 mm, or 0.9 mm, 1 mm, 2 mm, 3 mm, 4 mm, 5 mm, 6 mm, 7 mm, 8 mm, 9 mm, or 10 mm, or can be an amount falling within a range between any of the foregoing.

(136) It should be noted that, as used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the content and/or context clearly dictates otherwise. It should also be noted that the term “or” is generally employed in its sense including “and/or” unless the content or context clearly dictates otherwise.

(137) It should also be noted that, as used in this specification and the appended claims, the phrase “configured” describes a system, apparatus, or other structure that is constructed or configured to perform a particular task or adopt a particular configuration. The phrase “configured” can be used interchangeably with other similar phrases such as arranged and configured, constructed and arranged, constructed, manufactured and arranged, and the like.

(138) As used herein, the recitation of numerical ranges by endpoints shall include all numbers subsumed within that range, inclusive (e.g., 2 to 8 includes 2, 2.1, 2.8, 5.3, 7, 8, etc.).

(139) It is recognized that the figures shown and described are not necessarily drawn to scale, and that they are provided for ease of reference and understanding, and for relative positioning of the structures.

(140) The headings used herein are provided for consistency with suggestions under 37 CFR 1.77 or otherwise to provide organizational cues. These headings shall not be viewed to limit or characterize the invention(s) set out in any claims that may issue from this disclosure. As an example, a description of a technology in the “Background” is not an admission that technology is prior art to any invention(s) in this disclosure. Neither is the “Summary” or “Abstract” to be considered as a characterization of the invention(s) set forth in issued claims.

(141) The embodiments described herein are not intended to be exhaustive or to limit the invention to the precise forms disclosed in the following detailed description. Rather, the embodiments are chosen and described so that others skilled in the art can appreciate and understand the principles and practices. As such, aspects have been described with reference to various specific and preferred embodiments and techniques. However, it should be understood that many variations and modifications may be made while remaining within the spirit and scope herein.

(142) It is understood that although a number of different embodiments of the catheter systems have been illustrated and described herein, one or more features of any one embodiment can be combined with one or more features of one or more of the other embodiments, provided that such combination satisfies the intent of the present invention.

(143) While a number of exemplary aspects and embodiments of the catheter systems have been discussed above, those of skill in the art will recognize certain modifications, permutations, additions and sub-combinations thereof. It is therefore intended that the following appended claims and claims hereafter introduced are interpreted to include all such modifications, permutations,

additions and sub-combinations as are within their true spirit and scope, and no limitations are intended to the details of construction or design herein shown.

Claims

1. A catheter system for treating a treatment site within or adjacent to a blood vessel, the catheter system comprising: a power source; a light guide that receives power from the power source, the light guide having a distal tip, the light guide emitting light energy in a direction away from the distal tip; an inflatable balloon that encircles the distal tip of the light guide; and a plasma target that is spaced apart from the distal tip of the light guide by a target gap distance, the plasma target being positioned within the inflatable balloon, the plasma target being configured to receive light energy from the light guide so that a plasma is generated at the plasma target upon receiving the light energy from the light guide, the plasma target being at least partially formed from one of a polymer and a polymeric material.
2. The catheter system of claim 1 wherein the power source is a laser.
3. The catheter system of claim 1 wherein the light guide is an optical fiber.
4. The catheter system of claim 1 wherein the target gap distance is greater than 1 μm .
5. The catheter system of claim 1 wherein the target gap distance is greater than 100 μm .
6. The catheter system of claim 1 wherein the plasma target has one of a substantially circular, square, rectangular, oval, pentagonal, hexagonal, octagonal, polygonal, trapezoidal or diamond-shaped cross-sectional configuration.
7. The catheter system of claim 1 further comprising a guidewire lumen, the light guide being coupled to the guidewire lumen.
8. The catheter system of claim 1 wherein the plasma target has a target face that receives the light energy from the light guide, the target face being angled relative to a direction the light energy is emitted to the plasma target.
9. The catheter system of claim 1 wherein the plasma target is further partially formed from one of magnesium oxide, beryllium oxide, tungsten carbide, titanium nitride, titanium carbonitride and titanium carbide.
10. The catheter system of claim 1 wherein the plasma target is further partially formed from a ceramic material.
11. A catheter system for treating a treatment site within or adjacent to a blood vessel, the catheter system comprising: a power source; a light guide that receives power from the power source, the light guide having a distal tip, the light guide emitting light energy in a direction away from the distal tip; an inflatable balloon that encircles the distal tip of the light guide; and a plasma target that is secured to the light guide, the plasma target being positioned within the inflatable balloon, the plasma target being at least partially formed from (i) one of a polymer and a polymeric material, and (ii) one of tungsten, tantalum, platinum, molybdenum, niobium, and iridium, the plasma target including a target face that is spaced apart from the distal tip of the light guide by a target gap distance, the target face being configured to receive light energy from the light guide so that a plasma is generated at the target face upon receiving the light energy from the light guide, the target face being angled relative to a direction the light energy is emitted to the plasma target.
12. The catheter system of claim 11 wherein the power source is a laser.
13. The catheter system of claim 11 wherein the light guide is an optical fiber.
14. The catheter system of claim 11 wherein the target gap distance is greater than 1 μm .
15. The catheter system of claim 11 wherein the target gap distance is greater than 100 μm .
16. The catheter system of claim 11 wherein the plasma target has one of a substantially circular, square, rectangular, oval, pentagonal, hexagonal, octagonal, polygonal, trapezoidal or diamond-shaped cross-sectional configuration.
17. The catheter system of claim 11 further comprising a guidewire lumen, the light guide being

coupled to the guidewire lumen.

18. The catheter system of claim 11 wherein the plasma target is further partially formed from one of magnesium oxide, beryllium oxide, tungsten carbide, titanium nitride, titanium carbonitride and titanium carbide.

19. The catheter system of claim 11 wherein the plasma target is further partially formed from a ceramic material.

20. A catheter system for treating a treatment site within or adjacent to a blood vessel, the catheter system comprising: a power source; a light guide that receives power from the power source, the light guide having a distal tip, the light guide emitting light energy in a direction away from the distal tip; an inflatable balloon that encircles the distal tip of the light guide; and a plasma target that is spaced apart from the distal tip of the light guide by a target gap distance, the plasma target being positioned within the inflatable balloon, the plasma target being configured to receive light energy from the light guide so that a plasma is generated at the plasma target upon receiving the light energy from the light guide, the plasma target having a substantially circular cross-sectional shape, the plasma target being at least partially formed from (i) one of a polymer and a polymeric material, and (ii) one of tungsten, tantalum, platinum, molybdenum, niobium, and iridium.
